THE HONG KONG POLYTECHNIC UNIVERSITY
香港理工大學

BSE4610-

Building Informatics

Dr. Zhiling GUO (ZLG)

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Semester 1, 2026/27



Technical Foundations of Building Informatics

Week 2: Lectures + Tutorials (2.5 hours)

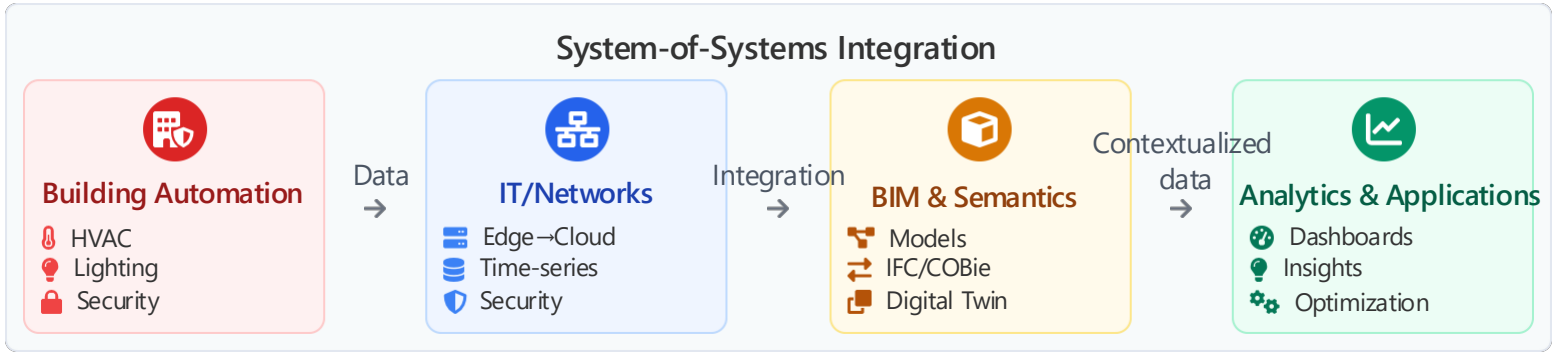
Focus today:

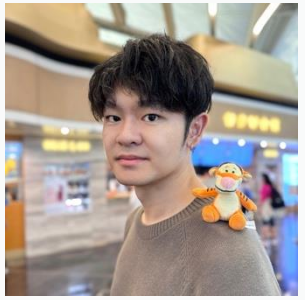
- Building Automation Systems (BAS)
- IT Infrastructure & Data Acquisition
- BIM for Building Informatics

Format:

Mini-lectures + quick polls + short activities

Building Informatics: A System-of-Systems View





Dr. Zhiling Guo (ZLG)

Department of Building Environment and Energy Engineering

Research interests:

- Machine Learning,
- Geoscience,
- AIoT,
- Renewable Energy Technologies.

Session Agenda and Time Plan

Part A: BAS fundamentals

45 min

- BAS architecture
- Core subsystems
- Control strategies
- Common protocols

Part B: IT & data

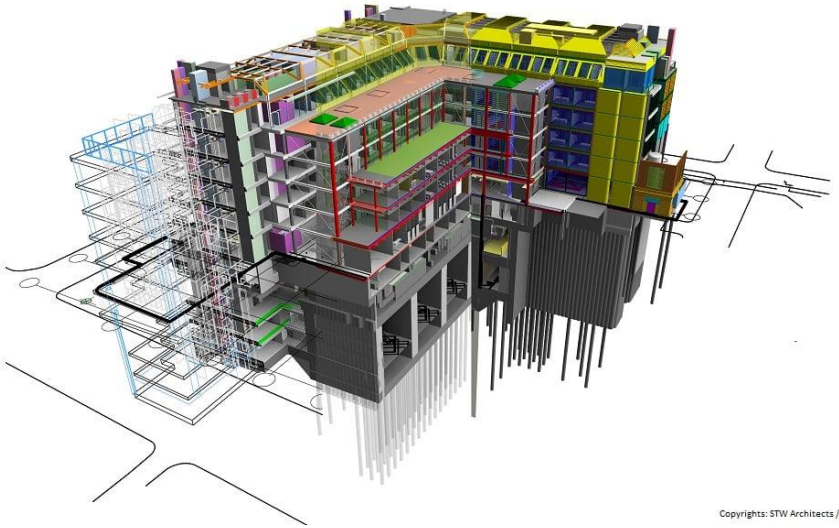
45 min

- Network infrastructure
- Edge→cloud architecture
- Time-series data
- Cybersecurity

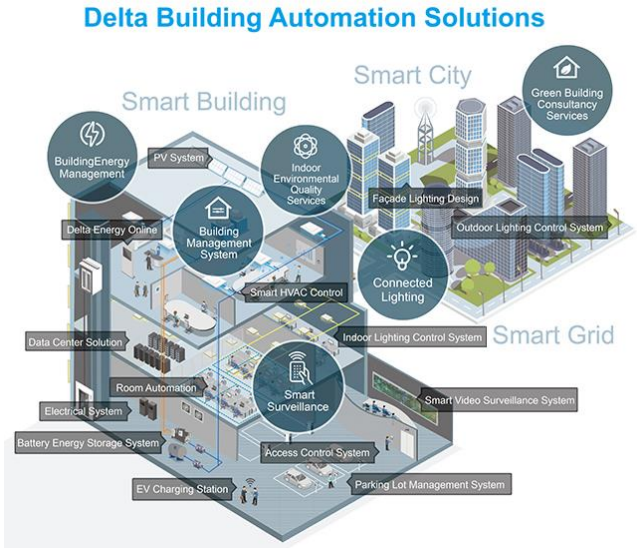
Part C: BIM & Integration

45 min

- Data-rich BIM
- IFC / COBie
- BIM-to-BAS mapping
- Digital twins



https://www.wsp.com/en-us/services/building-information-modelling-bim-in-healthcare



https://blog.amerlux.com/5-reasons-you-should-invest-in-a-building-automation-system/1000/

Activities

10 min

- Interactive polls
- Data tagging mini-task
- Short reflection

Q&A

5 min

- Open questions
- Next week preview

Learning Outcomes

By the end of this session, you can...

1 BAS Architecture

- Explain BAS architecture and control strategies

2 Building Protocols

- Identify common protocols and their roles

3 Cybersecurity

- Recognize key security controls for buildings

4 Data Acquisition

- Propose data pipeline (edge → time-series DB)

5 BIM Structure

- Describe BIM data and digital twin support

6 Semantic Tagging

- Map BAS points to enable analytics

Success Criteria - Measurable Skills:

Technical Sketch
You can sketch a control loop for an AHU and list monitored points

Data Management
You can outline how to timestamp, store, and query time-series data

Digital Twin
You can link BIM objects to real equipment and sensors

Session Agenda and Time Plan

Part A: BAS fundamentals

50 min

- BAS architecture
- Core subsystems
- Control logic strategies
- Common protocols

Part B: IT & data

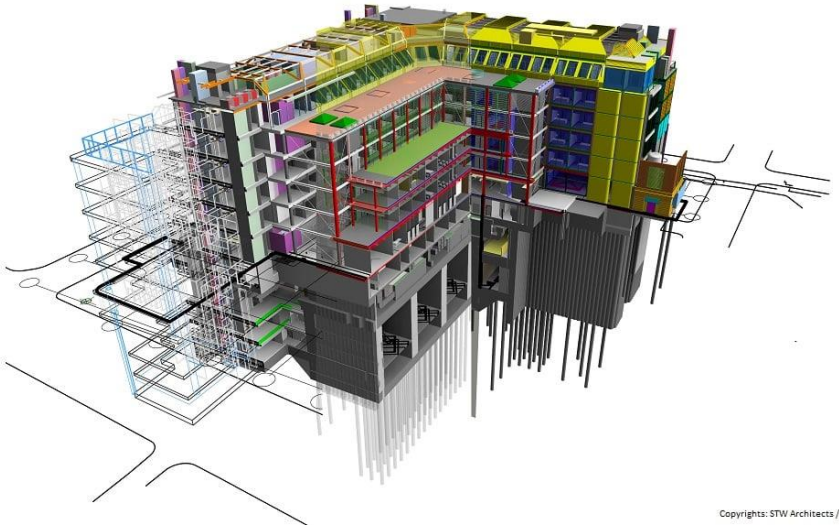
50 min

- Network infrastructure
- Edge→cloud architecture
- Time-series data
- Cybersecurity controls

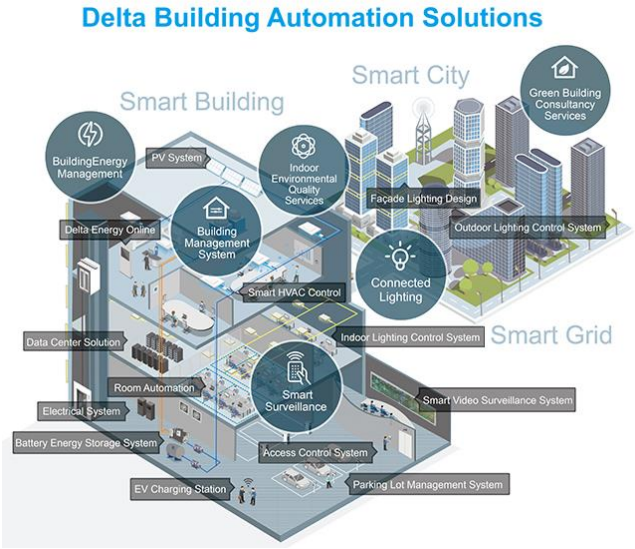
Part C: BIM

50 min

- Data-rich BIM models
- IFC/COBie standards
- Digital twin integration
- BIM-to-BAS mapping



https://www.wsp.com/en-us/services/building-information-modelling-bim-in-healthcare



https://blog.amerlux.com/5-reasons-you-should-invest-in-a-building-automation-system/1000/

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Building Automation Systems (BAS) Fundamentals

What Is a BAS?

Definition

Integrated monitoring and control of building systems (HVAC, lighting, security, vertical transport)

Goals

- Occupant comfort, safety
- Energy efficiency, reliability, maintainability

Core Components

Field Devices

- Sensors (temp, pressure)
- Actuators (valves, dampers)

Controllers

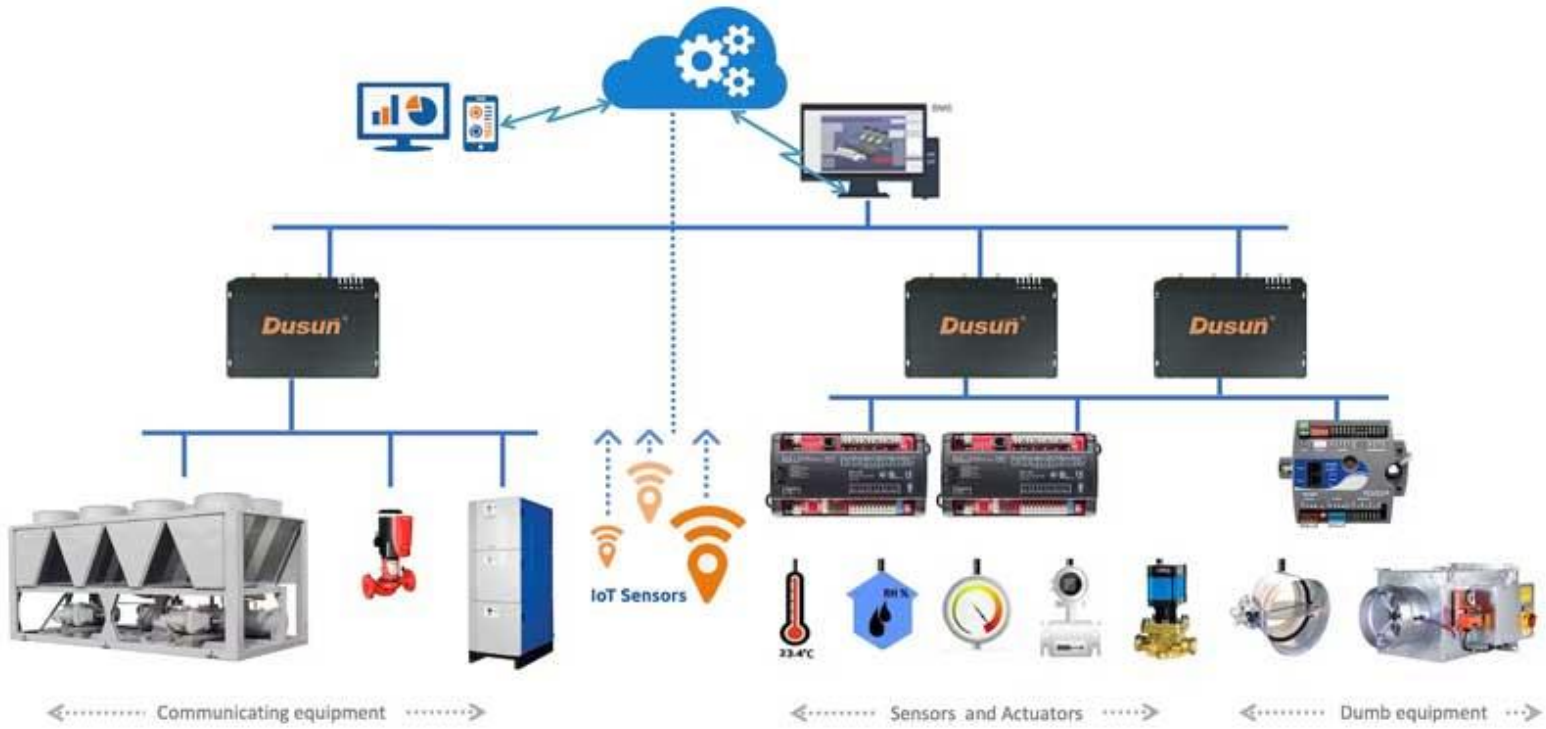
- DDC/PLC with control logic
- Local processing capability

Networks

- Fieldbus (BACnet MS/TP)
- IP networks (BACnet/IP)

Supervisory

- Servers, HMI/SCADA
- Alarms, trends, logs



Source: Dusun Inc.

Building Subsystems Overview

HVAC

- Central plant: chillers, boilers, cooling towers
- Air side: AHU, VAV/FCU, ducts, dampers

Security & Life Safety

- Access control, intrusion detection, fire alarms

Lighting

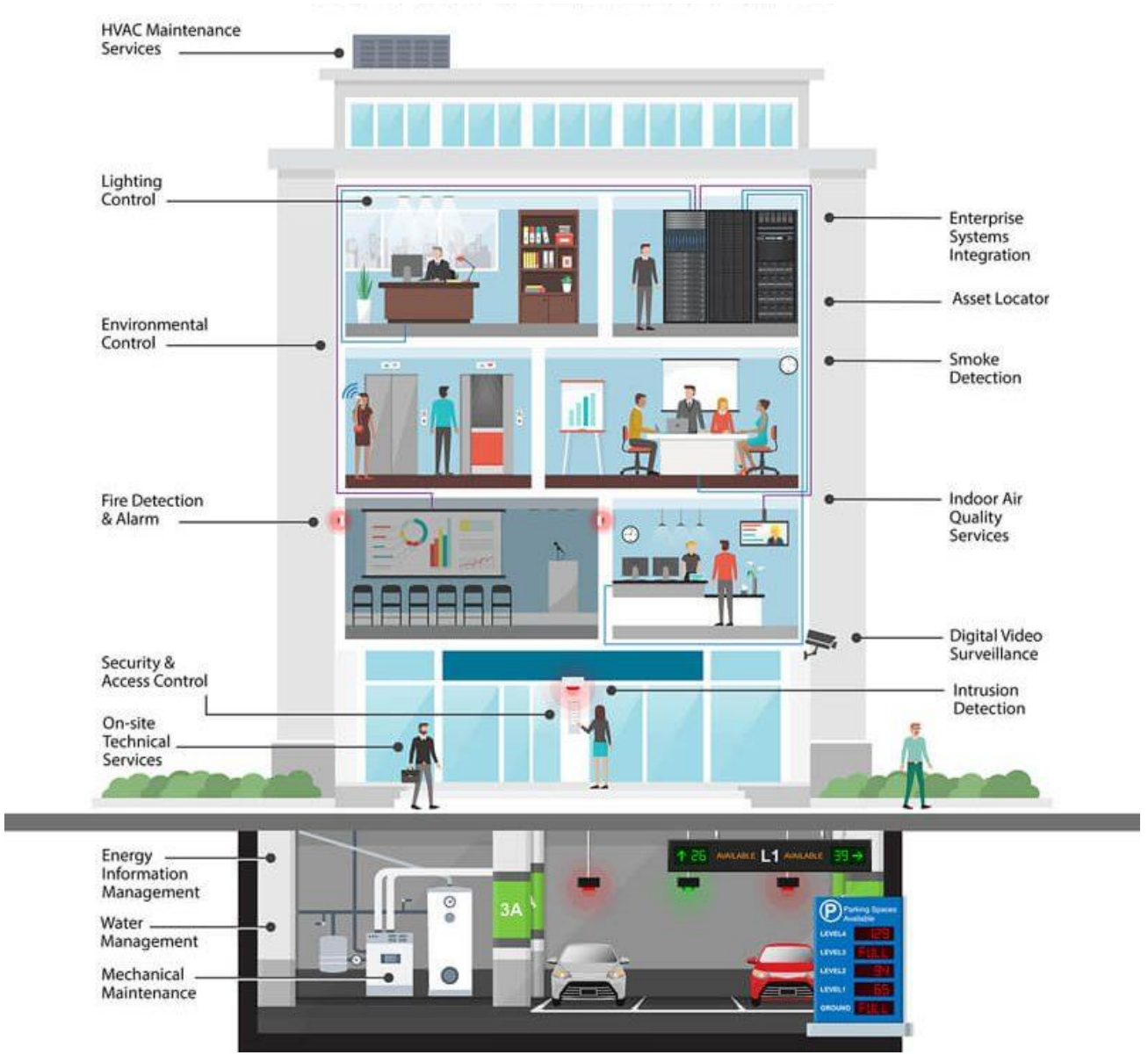
- Schedules, occupancy/daylight sensors, dimming

Vertical Transport

- Elevator group control, fire service integration

Interdependencies

- Demand response events affect HVAC/lighting
- Fire alarm systems may trigger predefined life-safety sequences that override normal HVAC operation



https://mes-teknologi.com/our-capabilities/building-automation-system/

HVAC Components

Understanding key equipment and parameters in building mechanical systems

Key Equipment

I. **Central Plant:** chillers, boilers, pumps, cooling towers
II. **AHU:** filters, coils, fans, dampers
III. **Terminals:** VAV boxes, FCUs

Typical Setpoints

- **Supply air:** 12-14°C
- **Duct static pressure:** ~250–500 Pa, , system-dependent
- **Chilled water:** 6-7°C
- **Zone temp:** 22-24°C

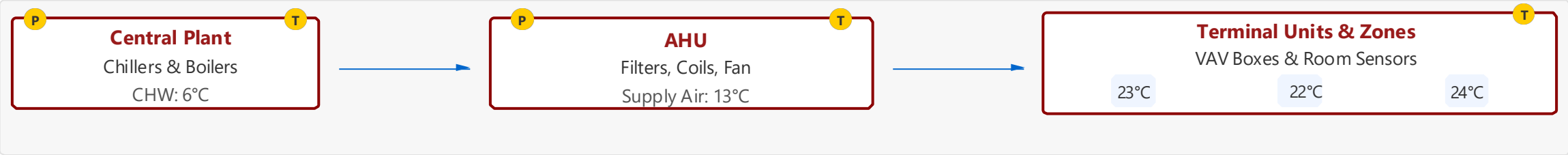
Sensors and Actuators

- **Sensors:** Temp, RH, CO2, pressure, flow
- **Actuators:** Valves, dampers
- **Controllers:** DDC (Direct Digital Control), PLC (Programmable Logic Controller)

Energy Levers

- **VFDs:** For fans and pumps
- **Economizer:** Free cooling
- **Reset strategies:** Demand-based setpoints

Simplified HVAC System Diagram



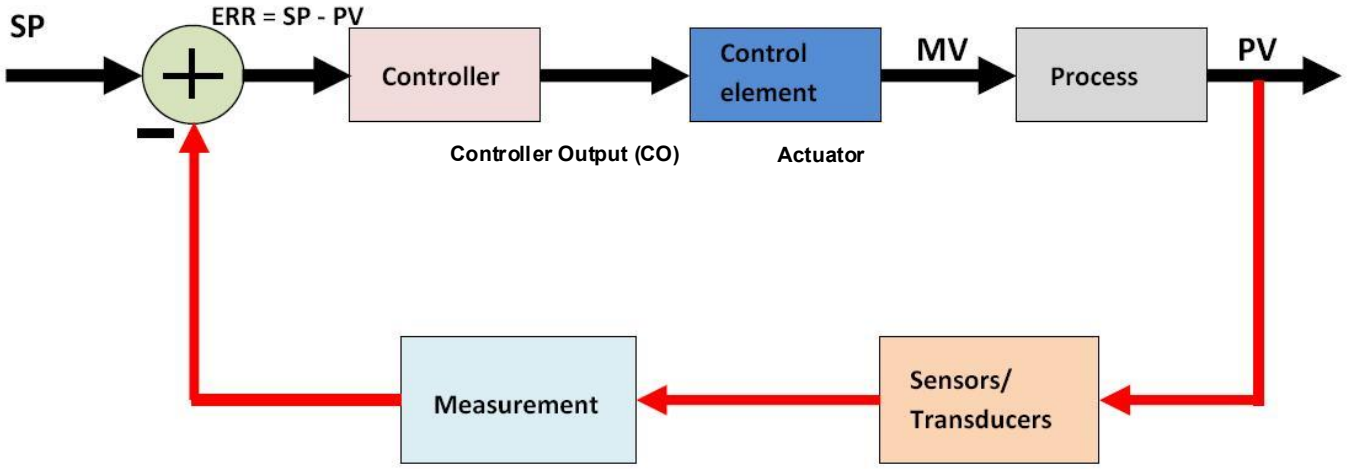
Legend: T = Temperature Sensor, P = Pressure Sensor

If the zone temperature sensor fails, how might the BAS respond incorrectly?

Control Loops Fundamentals

Understanding control theory for building systems

A Classic Control Loop Block Diagram:



How a Process Control Loop Works in Automatic Control Systems ~ Learning Instrumentation And Control Engineering

Core Elements

- Setpoint (SP): Desired value (e.g., 22°C)
- Process Variable (PV): Actual measured value
- Controller Output (CO): Signal to actuator (0-100%)

Loop Types

- Open loop: No feedback (e.g., timed lighting)
- Closed loop: Feedback-based correction
- Feedforward: Anticipates disturbances

Control Behavior

- Hysteresis: Prevents cycling (e.g., 0.5°C deadband)
- Local vs. supervisory: Point vs. system-level control

Typical HVAC Loops

- Zone temp → VAV damper position
- Supply air temp → cooling coil valve
- Static pressure → fan VFD speed

Loop Hierarchy Example:

Zone temperature rises → damper opens → pressure drops → fan speed increases → cooling valve modulates

AHU Control Strategies

Goals

- Maintain supply air temperature (SAT) and humidity
- Provide required ventilation with economizer when feasible
- Ensure safe operation (freeze, smoke, fire)

Core Loops

- SAT (supply-air-temperature) loop → chilled/hot-water valve modulation (PI)
- Supply fan static pressure loop → VFD (Variable Frequency Drive) speed

Economizer/Free Cooling

- Enable when OA enthalpy/temp < return air threshold
- High-limit shutoff (OA temp/enthalpy)
- Minimum OA damper position for ventilation

Humidity Control

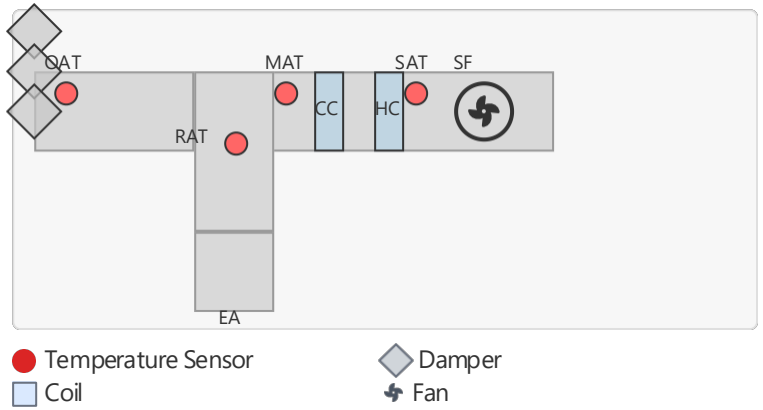
- Cooling for dehumidification, reheat as needed
- Low-limit SAT clamp to avoid coil freezing/condensation

Resets and Modes

- SAT reset based on warmest zone or VAV damper positions
- Night setback/standby/occupied modes
- Morning warm-up/cool-down sequences

OA = Outdoor Air
RA = Return Air
EA = Exhaust Air
MA = Mixed Air
SF = Supply Fan
SAT = Supply Air Temperature

AHU Schematic



Key Control Relationships:

- Economizer modulates OA/RA/EA dampers for free cooling
- Supply fan VFD maintains static pressure setpoint
- Valve positions adjust to maintain SAT setpoint

Lighting Control Strategies

Control Modes

- Scheduling/timeclocks with calendar exceptions
- Occupancy/vacancy sensors (auto-on to partial, auto-off)
- Daylight harvesting (closed-loop vs open-loop)
- Task tuning and scene control

Technologies

- DALI/DALI-2 networks; 0–10 V analog; DMX for specialty
- Power over Ethernet (PoE) lighting systems
- Sensors: PIR, ultrasonic, dual-tech; photosensors

Integration

- BMS interface for status, overrides, demand response
- Emergency lighting testing/reporting interfaces

Lighting Zones & Sensor Placement



Design Considerations

- Zoning, sensor placement/height, tuning thresholds
- Glare, flicker (IEEE 1789), color quality (CRI/CCT)
- Tunable white lighting for circadian wellness

Energy & Codes

- Automatic shutoff, partial on, daylight zones
- Measurement and verification via trend logs
- Compliance with ASHRAE 90.1, LEED requirements

Protocol Comparison

Protocol	Addressability	Wiring	Feedback
DALI/DALI-2	High (64 devices)	2-wire bus	Yes
0-10V	Limited	2-wire analog	No
PoE Lighting	High (IP)	CAT5/6	Yes

Typical Energy Savings

Scheduling	Occupancy	Daylighting
10-15%	20-30%	15-40%

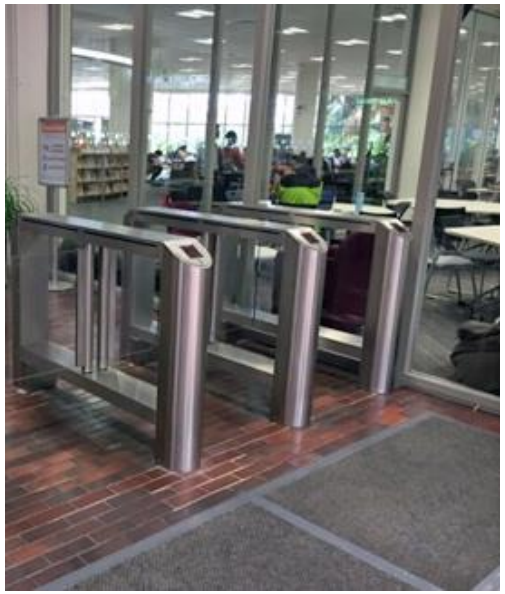
Security & Access Control Integration

Components

- Credentials (cards, mobile, biometrics), readers
- Door controllers and intelligent panels
- Locks: fail-safe vs fail-secure; position sensors
- Request-to-exit (REX) devices and motion detectors

Integrations

- BMS: occupancy signals, after-hours HVAC enable
- Video management (VMS): event-linked recording
- Elevator integration for floor-specific permissions
- Visitor management and pre-registration systems



Core Functions

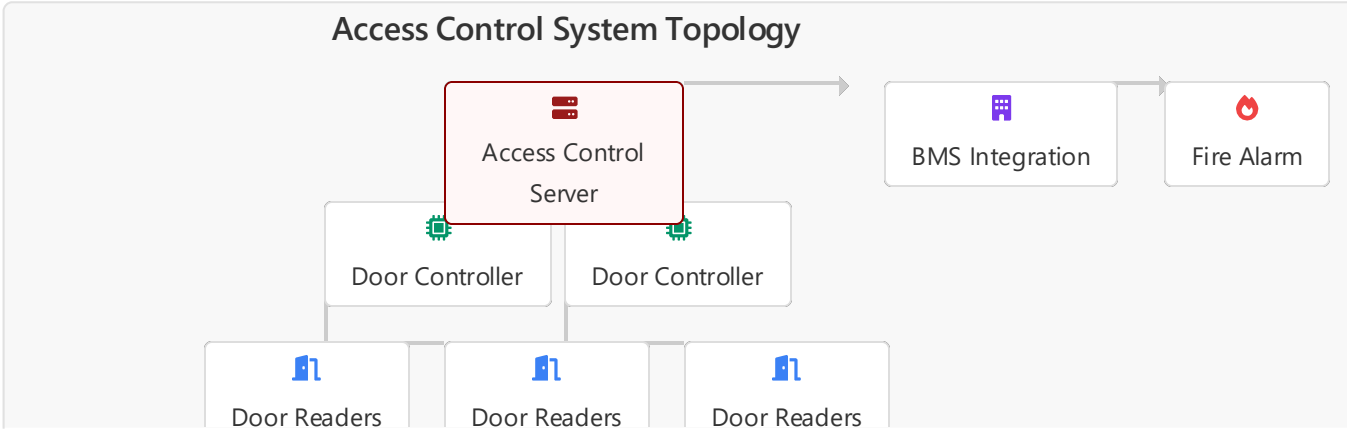
- Access rules (schedules, roles, zones)
- Anti-passback logic to prevent credential sharing
- Alarm events: door forced/held open, tamper detection
- Real-time monitoring and muster reports

Safety & Compliance

- Fire alarm interface: emergency door release paths
- Egress requirements and life safety regulations
- Privacy and data retention policies (PII handling)
- Audit trails for compliance reporting

Cybersecurity

- Encrypted readers/credential tech (OSDP Secure)
- Mobile credentials using BLE/NFC encryption
- Network segmentation and access control
- Vendor remote access and firmware updates



BAS Devices & Architecture

BAS Architecture Layers

Integration/Enterprise Layer

- APIs, REST/GraphQL interfaces
- MQTT/OPC UA message brokers
- CMMS/EMIS/BIM/Digital Twin integration
- Enterprise dashboards and business intelligence

Management Layer (Supervisory)

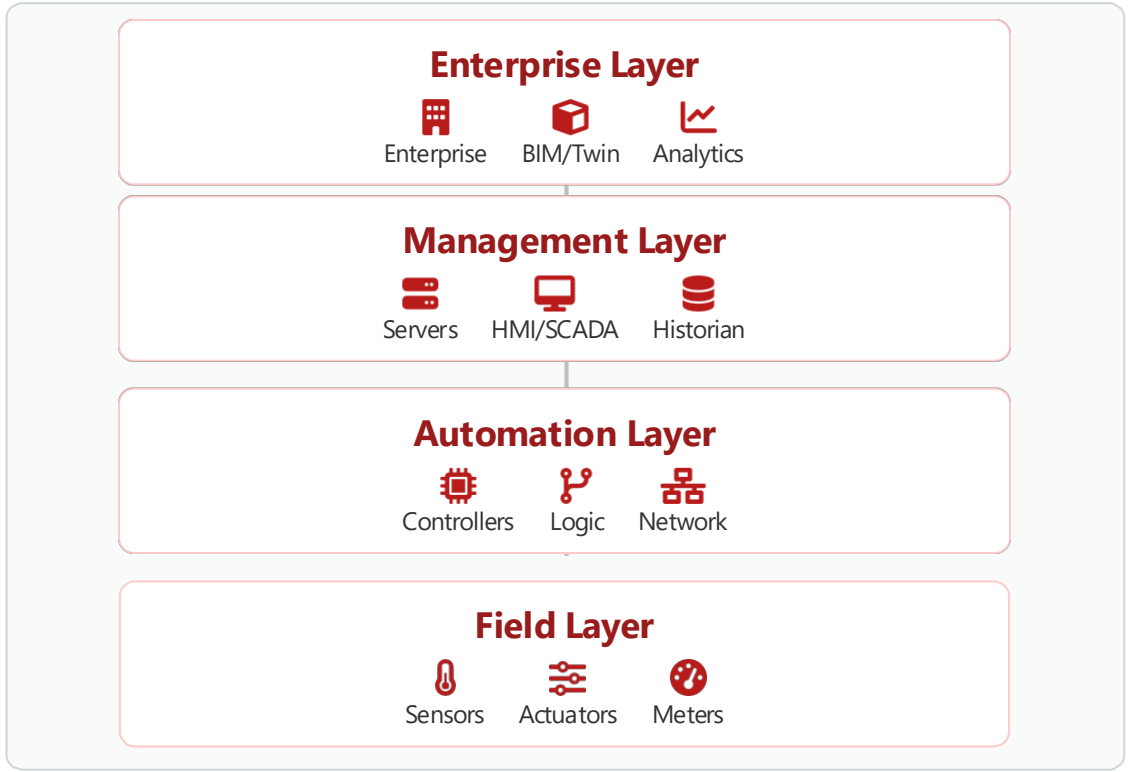
- Servers, HMIs, SCADA, visualization
- Historians, analytics, trend logging
- Integration gateways and protocol converters
- Global scheduling, alarms and notifications

Automation Layer (Control)

- DDC/PLC controllers running local control logic
- Sequence of Operations (SoO) implementation
- BACnet/IP, BACnet MS/TP, Modbus networks
- Control loops, interlocks, scheduling

Field Layer (I/O)

- Sensors: temperature, pressure, CO_2, flow
- Actuators: valves, dampers, VFDs
- Meters: electrical, BTU, water, gas



Cross-cutting Concerns:

Time Sync (NTP) | Security (Auth/Seg) | Redundancy

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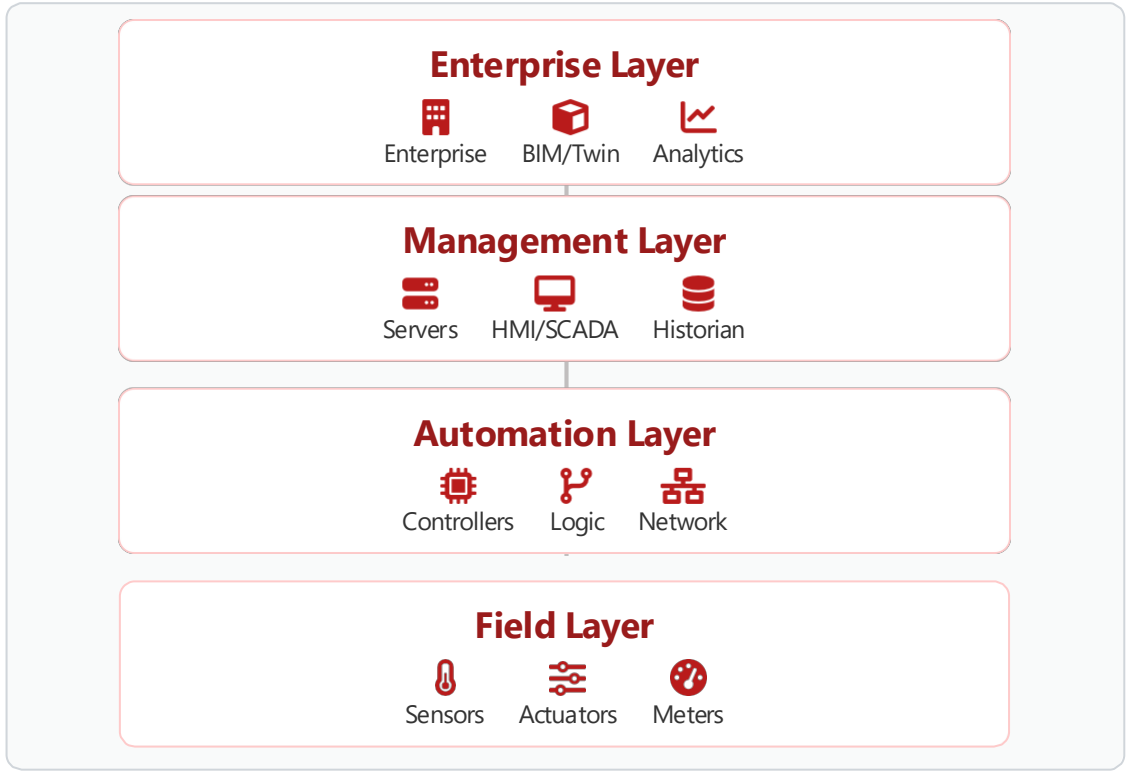
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Sensors: Types, Principles, and Selection

Environmental Sensors

- Temperature (RTD Pt100/1000, thermistor NTC), RH (capacitive), dew point calculation
- CO_2 (NDIR), VOC (MOX)
- PM2.5 (laser scattering)
- Pressure (differential for filters/ducts)
- Flow (Pitot tubes, ultrasonic, vortex)

Electrical/Energy Sensors

- Power meters (true RMS, harmonics measurement)
- Pulse outputs (kWh, water, gas)
- Current transformer types (split-core/solid)
- Phase configurations (3-phase, single phase)
- Accuracy classes (0.5S, 1.0, 2.0)

Selection Criteria

- Accuracy, range, resolution, repeatability
- Stability/drift characteristics over time
- Response time and sampling frequency
- Environmental rating (IP/NEMA)
- Calibration requirements and field adjustability

Placement Best Practices

- Avoid radiant bias for temperature sensors
- Representative airstream locations (straight runs)
- RH/CO_2 at breathing zone height (1.1-1.7m)
- PM sensors away from direct drafts
- DP taps per manufacturer guidance

Data Quality

- Acceptance checks (range, rate-of-change)
- Periodic calibration schedule and verification
- Reference sensor cross-checking
- Statistical outlier detection
- Maintenance logs and sensor history



https://www.designingbuildings.co.uk/wiki/Types_of_building_sensors

RTD (Pt100) | RH Sensor | Pressure | CO_2 (NDIR) | CT (Split-core) | Flow Meter

BAS Point Types and Semantics

Point Categories

AI/BI
Analog/Binary Inputs
Sensors, Statuses

AO/BO
Analog/Binary Outputs
Actuators, Commands

AV/BV
Software Values
Setpoints, Mode Flags

Typical Examples

Point Type	Meaning	Example Functions
AI (Analog Input)	Sensor input with continuously variable values	Temperature sensors, voltage, current readings
AO (Analog Output)	Output with variable control signals	Controlling fan speed, valve position, dimmable lighting
BI (Binary Input)	Binary (on/off) input signals	Door open/closed, equipment running status
BO (Binary Output)	Binary (on/off) control signals	Turning fans or lights on/off
BV (Binary Value)	Logic-based binary state	Used for internal logic or mode switching (e.g., On/Off, Auto)
AV (Analog Value)	Adjustable/programmable analog values	Setpoints for temperature, thresholds, control parameters
Schedule	Time-based control point	Weekly timers for HVAC, lighting, system activation

Engineering Details

- Units and ranges: °C, Pa, ppm, L/s with proper scaling
- Enumeration for binary states (0=Closed, 1=Open)
- Change-of-value (COV) reporting vs periodic polling
- Offset and scaling: raw value to engineering units
- Linearization for non-linear sensors (thermistors)

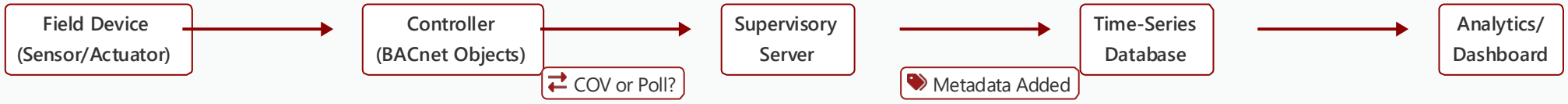
Quality and Flags

- Status indicators: reliable/unreliable, overridden
- Alarm limits: high/low, rate-of-change, delay
- Timestamping: local vs server time
- Out-of-service flag for maintenance

Naming Conventions

- Consistent, hierarchical, machine-readable
- Format: [Building].[System].[Equipment].[Point].[Type]
- Example: PolyU.HVAC.AHU1.SAT.AI
- Include units in metadata: degC, Pa, %

Data Flow Through BAS



Network Topologies in BAS

Physical Media Options

- **RS-485:** MS/TP (BACnet), Modbus RTU
- **Ethernet:** BACnet/IP, Modbus TCP
- **Wireless:** Wi-Fi, Zigbee, BLE, LoRaWAN
- Media selection factors: distance, noise immunity, cost

Network Topologies

- **Daisy chain/bus:** Common for RS-485, simple wiring
- **Star:** Ethernet switches, point-to-point connections
- **Ring:** Resiliency through redundant paths (RSTP)
- **Hybrid:** Mixed topologies in larger systems

RS-485 Best Practices

- Single continuous bus with 120Ω termination at both ends
- Biasing resistors to stabilize idle voltage levels
- Avoid stubs (<0.3m) to prevent signal reflections
- Max 32 nodes at 9600 baud (~1200m max length)
- Shielded twisted pair in noisy environments

Practical knowledge

Ethernet Best Practices

- Managed switches with VLAN segregation
- RSTP/MSTP for loop prevention and redundancy
- PoE budgets for sensors, wireless APs, displays
- 100m max copper run; fiber for longer distances
- Network segmentation with ACLs for security

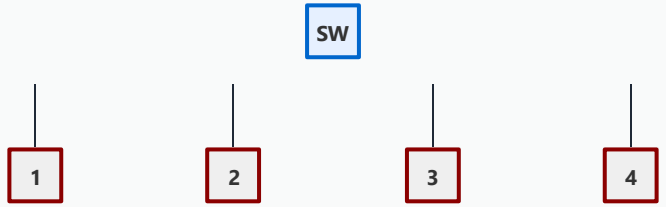
Network Topology Comparison:

Daisy Chain/Bus (RS-485)



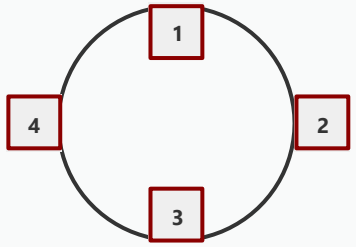
Single communication path; termination at ends

Star (Ethernet)



Central switch with point-to-point connections

Ring (Redundant Ethernet)



Fault tolerance with RSTP/MSTP protocols

Communication Protocols in BAS

BAS Architecture Layers

Integration/Enterprise Layer

- APIs, REST/GraphQL interfaces
- MQTT/OPC UA message brokers
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- Enterprise dashboards and business intelligence

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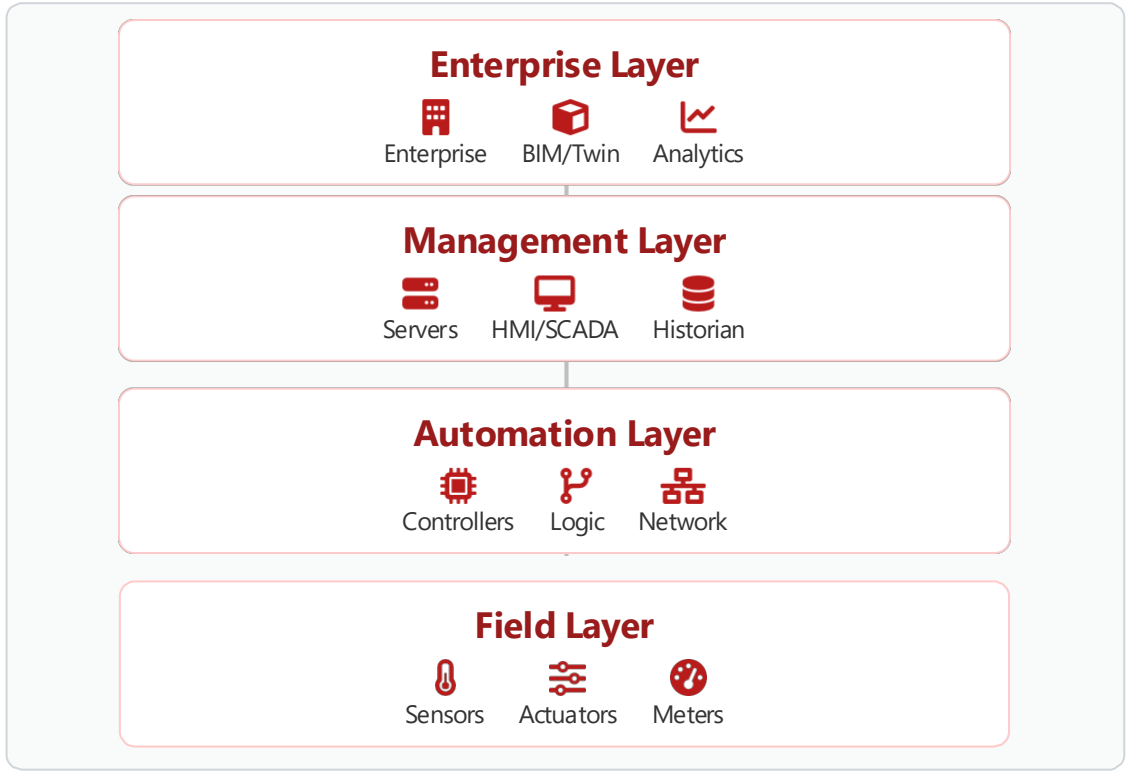
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Cross-cutting Concerns:

Time Sync (NTP) | Security (Auth/Seg) | Redundancy

Communication Protocols Landscape

Major Protocol Families

BACnet Modbus LonWorks KNX

- **BACnet (ASHRAE 135):** Objects/services model for buildings HVAC
- **Modbus:** Simple register/coil model, RTU/TCP variants
- **LonWorks:** LonTalk protocol with standardized profiles
- **KNX:** Lighting/shading dominant; TP/IP/RF media

Open vs Proprietary

- **Open standards** ease multi-vendor interoperability
- Documented specifications and independent certification
- Beware "open protocol" claims with proprietary extensions
- Consider vendor ecosystem and local support availability

Selection Guidelines

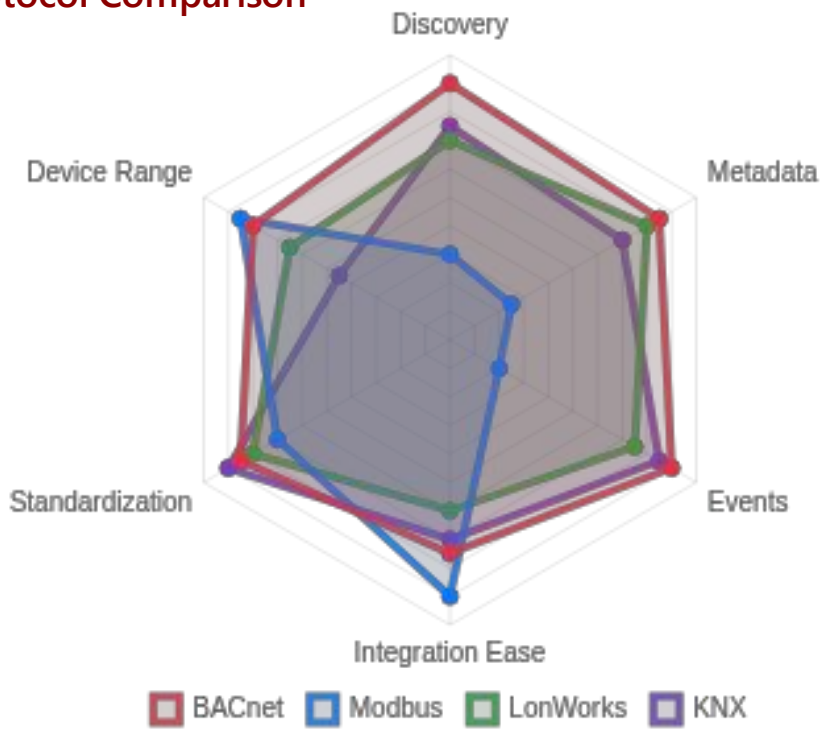
Rich metadata & events?
BACnet, OPC UA

Simple sensor polling?
Modbus RTU/TCP

Lighting & room controls?
KNX, DALI

Legacy MEP systems?
LonWorks, BACnet MS/TP

Protocol Comparison



Gateways & Translation

- **Data loss risks:** Semantic loss, scaling/unit errors, latency
- **Performance issues:** Polling bottlenecks, buffer overflows
- **Reliability:** Single point of failure, fault monitoring needed
- **Best practice:** Document mappings, monitor health, plan redundancy

BAS Operations & Interfaces

Graphics & HMI Design

Creating effective building automation interfaces

Usability Principles

- Consistency and clarity
- Minimal cognitive load
- Color coding by standards
- Show state, not just data

Navigation Hierarchy

- Campus → Building → System → Equipment → Point level
- Breadcrumbs & quick links
- Contextual navigation

Data Presentation

- Sparkline trends near values
- Contextual limits & targets
- Units & quality flags
- Time since last update

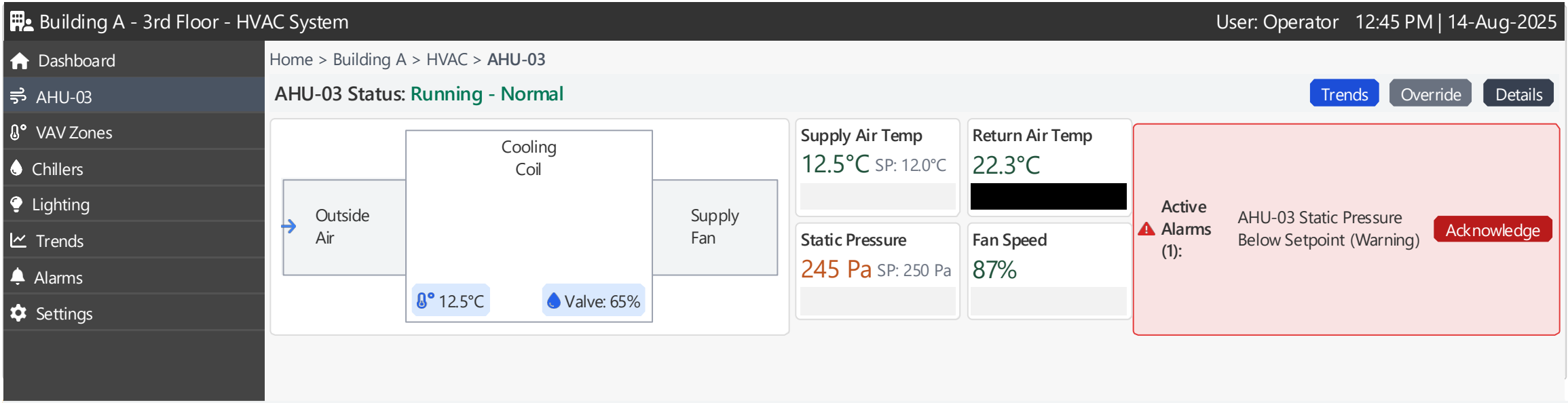
Accessibility

- Responsive layouts
- 10-12 ft viewing distance
- Keyboard navigation
- High-contrast mode
- Multi-language support

Alarm Visualization

- Consolidated alarm banner
- Severity & filtering
- Root cause drill-down
- Related points context

Mock HMI Screen: AHU Interface



Intro to Fault Detection & Diagnostics (FDD)

Automated detection and diagnosis of building system issues

What is FDD?

- Automated detection of abnormal conditions and identification of likely causes
- Proactive maintenance approach that reduces downtime and energy waste
- Integration of engineering knowledge, data analytics, and operational feedback

Approaches

- **Rule-based:** Physics/SoO rules, thresholds, persistence checks
- **Model-based:** First principles or data-driven residuals analysis
- **Data science:** Supervised classification, unsupervised anomaly detection

HVAC Examples

Stuck Cooling Valve
Low SAT response vs high valve command with stable load

Sensor Bias
MAT ≠ mix(OAT,RAT,damper pos) by tolerance

Low ΔT Syndrome
High flow + low CHW ΔT for sustained period

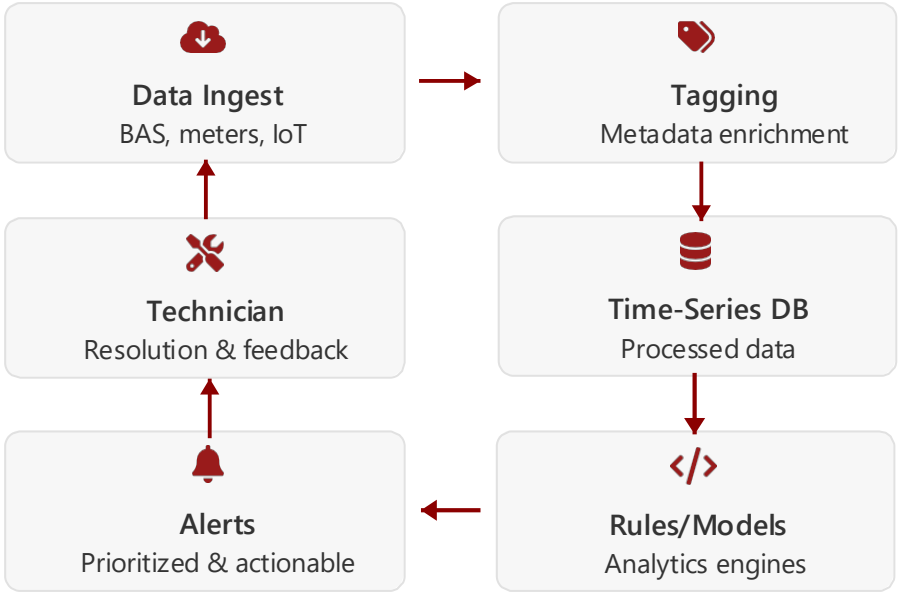
Data Requirements

- Synchronized time-series data with consistent sampling intervals
- Complete metadata and tags to interpret points and relationships
- Quality flags to filter unreliable sensor readings
- Baselines and seasonal models for context-aware analytics

Design for Actionability

- **Severity scoring:** Impact on energy, comfort, equipment life
- **Persistence:** Filter out transients, confirm real issues
- **Confidence metrics:** Avoid false positives that erode trust
- **Maintenance links:** Connect to SoO, manuals, parts lists
- **CMMS integration:** Auto-generate work orders with context

FDD Workflow



Session Agenda and Time Plan

Part A: BAS fundamentals

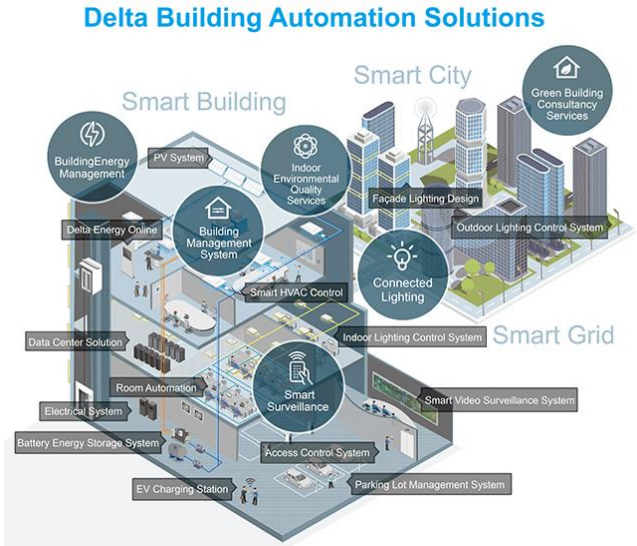
50 min

- BAS architecture
- Core subsystems
- Control logic strategies
- Common protocols

Part B: IT & data

50 min

- Network infrastructure
- Edge→cloud architecture
- Time-series data
- Cybersecurity controls



https://blog.amerlux.com/5-reasons-you-should-invest-in-a-building-automation-system/1000/

Part C: BIM

50 min

- Data-rich BIM models
- IFC/COBie standards
- Digital twin integration
- BIM-to-BAS mapping



https://www.wsp.com/en-us/services/building-information-modelling-bim-in-healthcare

Activities

10 min

- Interactive polls
- Data tagging mini-task
- Short reflection

Q&A

5 min

- Open questions
- Next week preview

IT & Networking for Smart Buildings

IT/OT Convergence

Different Priorities

	OT (Operational Technology)	IT (Information Technology)
Focus	Availability, safety, reliability	Confidentiality, integrity, agility
Lifecycle	Long term (15-20 years)	Short term (3-5 years)
Control	Deterministic response	Scalability, flexibility

Governance & Processes

- Change control windows, patch cycles, rollback plans
- Asset inventory (CMDB), configuration baselines
- Regular backups and disaster recovery testing
- Joint incident response procedures

Architecture Patterns

- Segmented networks with clear boundaries
- OT DMZ for secure data exchange via brokers
- API-first integrations over direct database access
- Defense-in-depth: multiple security layers

In-Class Prompt:

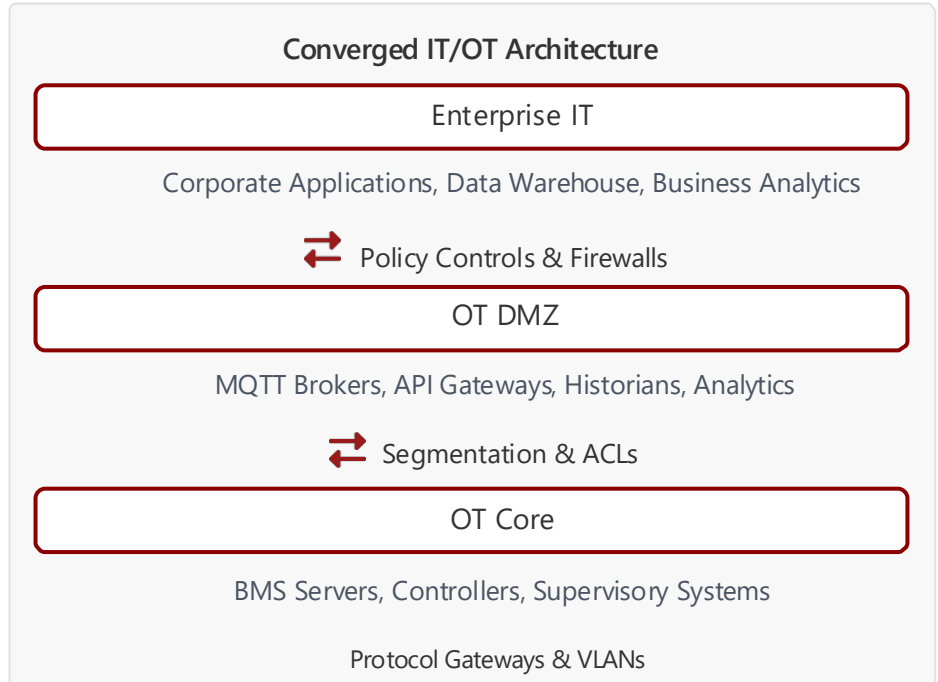
What could go wrong if an HVAC controller is exposed on the open internet?

Collaboration Model

- RACI matrix: clearly defined responsibilities
- Joint security operations center (SOC)
- Cross-functional teams for projects
- Regular tabletop exercises and simulations

Remote Access & Vendors

- Multi-factor authentication (MFA)
- Jump hosts and session recording
- Least-privilege vendor accounts
- Contractual SLAs for patching and vulnerabilities

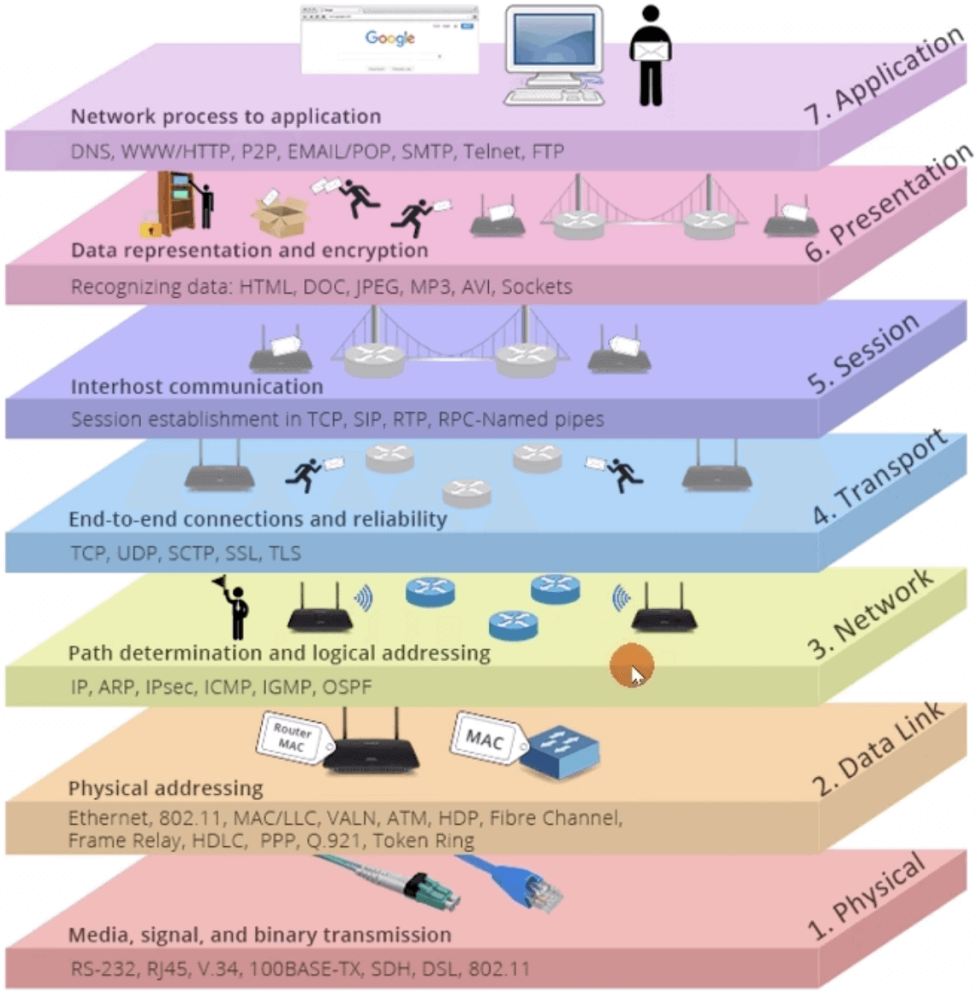


Network Models: OSI & TCP/IP (for BAS)

Understanding network layers in Building Automation Systems

OSI vs TCP/IP Network Models (with Protocols & BAS Practical Mapping)

OSI (Open Systems Interconnection) Layer	TCP/IP (Transmission Control Protocol / Internet Protocol) Layer	Typical Protocols / Functions	BAS Practical Mapping
7. Application	Application	HTTP, FTP, TFTP, NFS, WAIS, SMTP	**BACnet/IP**, **Modbus TCP**, **MQTT**, **OPC UA**, RES APIs
6. Presentation	Application	Data representation / encryption functions	**TLS encryption**, **JSON**, **CBOR**, XML for data formatting
5. Session	Application	Session management functions	**TLS handshake**, **MQTT session management**, persistent device connections
4. Transport	Transport	TCP, UDP	**TCP**: reliable byte-stream transport — e.g., Modbus TCP, MQTT **UDP**: connectionless datagrams — e.g., classic BACnet/IP, NTP, many discovery functions
3. Internet	Network	IPv4/IPv6, ICMP	**IP addressing**, **subnetting**, **routing**, **BACnet BBMDs** for broadcast across subnets
2. Data Link	Network Access	Ethernet, FDDI, Arpanet, SLIP, PPP	**Ethernet (802.3)**, **Wi-Fi (802.11)**, **VLAN (802.1Q)**
1. Physical	Network Access	Physical media/signaling: copper, fiber, radio; Ethernet PHY, Wi-Fi PHY, RS-485 electrical layer	**Twisted-pair cables**, **Fiber optics**, **RS-485**, **Radio Frequency (RF)** for wireless BAS devices



https://javaguide.cn/cs-basics/network/osi-and-tcp-ip-model.html#osi-%E4%B8%83%E5%B1%82%E6%A8%A1%E5%9E%8B

Wired vs Wireless in Buildings

Wired (Ethernet/fiber)

- Pro Reliability, low latency/jitter, PoE power, security
- Con Higher install cost, less flexibility
- Cabling: Cat6/6A for 1/2.5/10 GbE; fiber (SM/MM) for backbone

Wireless (Wi-Fi/LP-WAN/personal area)

- Pro Flexibility, retrofit-friendly, battery devices
- Con Interference, coverage gaps, battery maintenance
- Options: Wi-Fi, Zigbee, Thread, BLE, Z-Wave, LoRaWAN

Selection Guidance

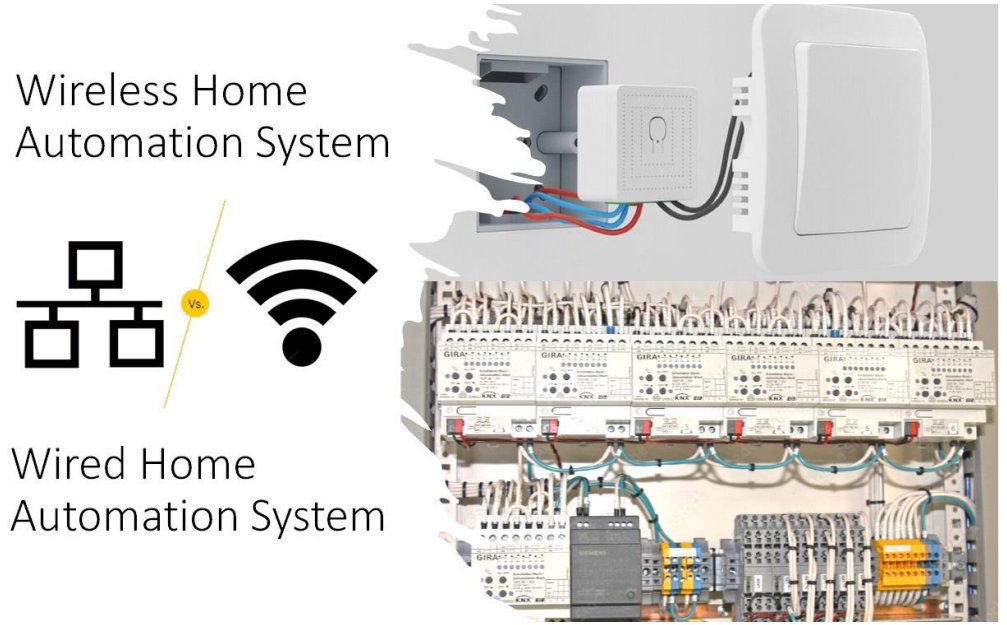
- Critical control loops → wired; noncritical telemetry → wireless okay
- High-density sensors → mesh (Zigbee/Thread); long range → LoRaWAN

Design & Operations

- Site surveys, spectrum analysis; channel planning
- PoE and AP placement; antenna selection; enclosure impacts
- Battery life modeling (reporting intervals, sleep modes)

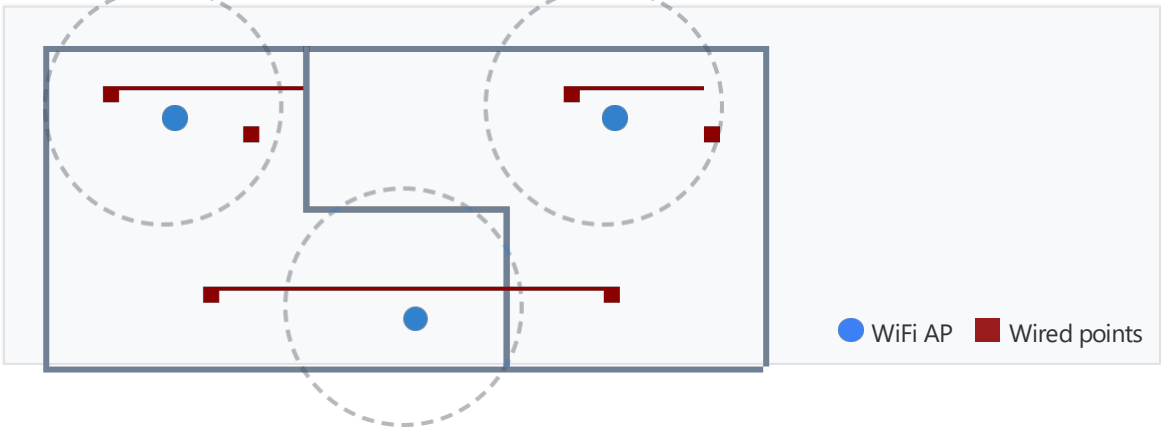
Technology Comparison Matrix

Technology	Range	Throughput	Latency	Power	Use Case
Ethernet	100m	High	Very Low	PoE	Controllers, HMI
Wi-Fi	50-100m	Medium-High	Low	Medium	Mobile access, video
Zigbee/Thread	10-30m	Low	Medium	Very Low	Room sensors, switches
LoRaWAN	2-15km	Very Low	High	Ultra Low	Utility meters, parking



https://www.knxhub.com/wired-vs-wireless-home-automation/

Building Floor Plan with Network Infrastructure



Cloud & Data Infrastructure

Cloud Architectures for Building Analytics

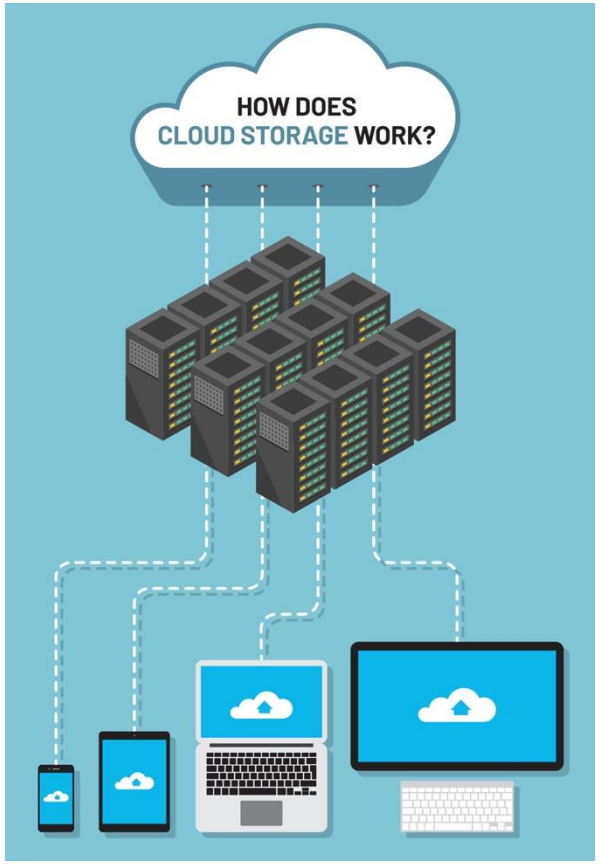
Reference Layers & Data Pipelines

Ingest
- MQTT Broker
- IoT Hub
- API Gateway

Stream/Process
- Lambda Functions
- Stream Processors

Storage
- Time Series DB
- Data Lake
- Data Warehouse

Serve
- APIs
- Dashboards
- ML Services



Source: Cloud Data Storage Types Top-notch - Cloud Ready Storage Quiz

Data Pipelines: Ingest → Store → Serve

Ingest → Transform → Store → Serve → Governance

Time Series Data Characteristics (in Buildings)

Understanding the unique nature of building data for effective analytics

Statistical Properties

- Regular vs irregular/event-driven sampling patterns
- Diurnal & weekly seasonality in occupancy and energy use
- Non-stationarity and regime changes (seasons, schedules)
- Autocorrelation within points; cross-correlation between systems

Engineering Realities

- Clock drift and timestamp skew between edge and server
- DST transitions and timezone challenges in global portfolios
- Missing data, gaps, spikes from sensor/network issues
- Sensor noise, quantization, and calibration drift over time

Storage/Query Implications

- Schema design: narrow vs wide tables; time+entity indexing
- Compression techniques (Gorilla, delta-of-delta encoding)
- High-cardinality tag management for building metadata
- Rollups and downsampling strategies with preservation of extremes

Analytics Readiness

- Resampling, interpolation methods with quality flags
- Outlier detection and cleaning techniques for building data
- Windowing strategies: tumbling, sliding, session windows
- Feature engineering: lags, derivatives, integrals for ML

Sampling, Resolution, and Latency

Choose rates by process dynamics

- Fast loops (fan VFD, valve pos): 1–5 s
- Air temps/pressures: 5–15 s
- Energy (kW): 1–15 s; kWh rollups: 1–15 min
- Occupancy/events: event-driven or 5–30 s heartbeat

Aliasing and fidelity

- Nyquist: sample ≥2× highest expected signal frequency
- Use anti-alias filters or longer windows for noisy signals

Event vs time-based collection

- COV for slow-changing points with periodic heartbeats
- Hybrid: COV for small Δ, periodic snapshots for freshness

Recommended Sampling Rates:

Point Type	Sample Rate	Approach
Critical Control	1-5 seconds	Time-based
Temperature	10-15 seconds	COV + 60s heartbeat
Electrical Load	5-15 seconds	Time-based
Occupancy	Change + 30s	Event-driven

Resolution and quantization

- Sensor precision and A/D bits matter (e.g., 0.1 °C vs 0.01 °C)
- Avoid over-precision that exceeds sensor accuracy

End-to-end latency budgets

- Sensor → controller → gateway → broker → TSDB → app
- Define targets (e.g., ops dashboard ≤5 s; FDD ≤60 s)

Jitter management

- Buffering and clock sync to align windows
- Rate limiting during bursts

End-to-End Latency Budget:

Sensor & Edge	Network	Ingest/MQTT	TSDB Write	Query Cache	Total
500ms	1000ms	500ms	500ms	1000ms	3.5s

Dashboard target: ≤5s | Control loops: ≤2s | Analytics/FDD: ≤60s

Time Synchronization

Accurate time across building systems

Protocols and Roles

- NTP for general sync; local stratum-1/2 servers
- PTP (IEEE 1588) for sub-millisecond precision

Architecture

- One authoritative time source per site
- Boundary/transparent clocks on PTP switches

Timestamps and Storage

- Store in UTC; display in local time
- Capture both device and server timestamps

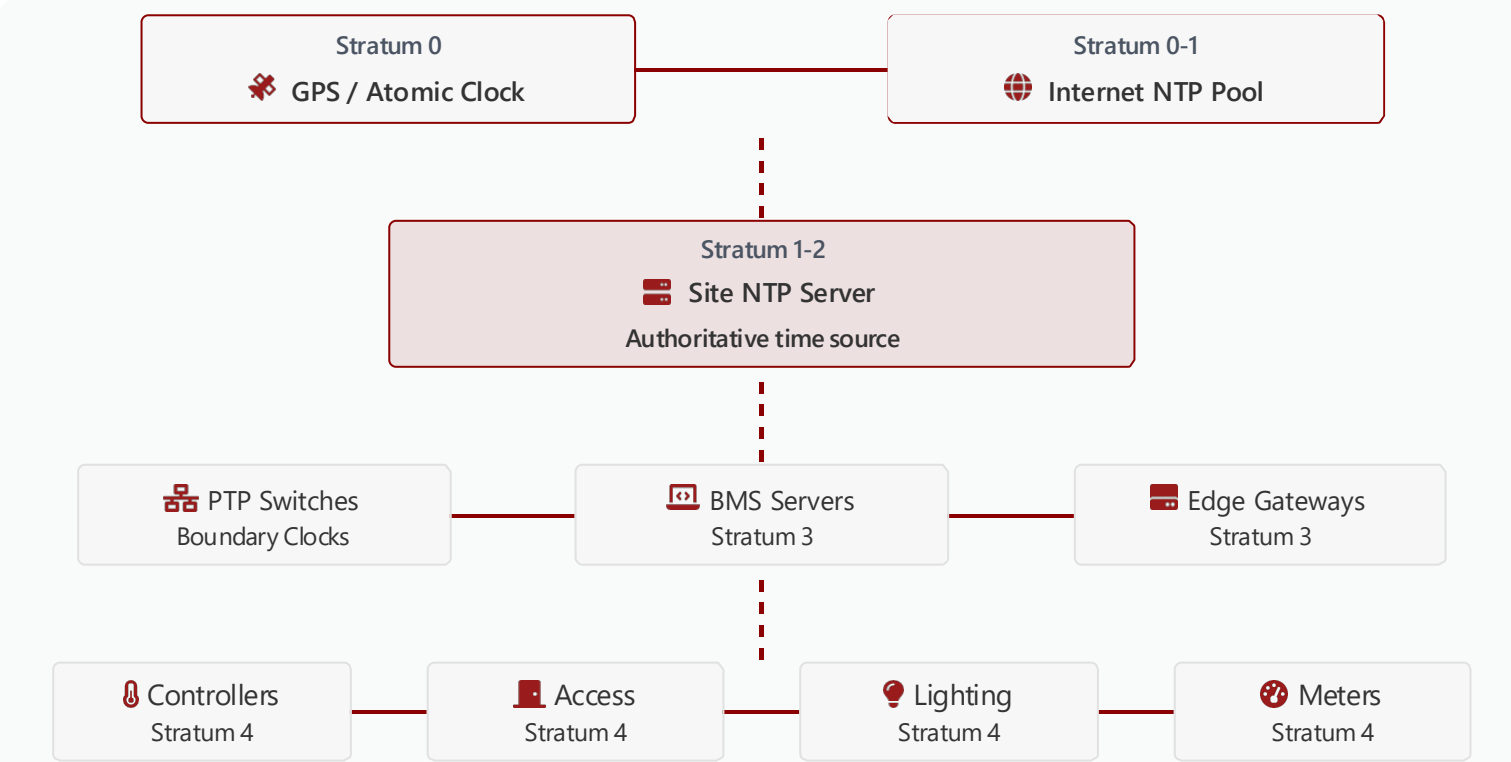
Clock Health

- Drift monitoring and alarms
- Max skew SLA (e.g., ≤200 ms for BAS)
- Secure NTP (NTS or keyed authentication)

Edge Cases

- DST: handle fall-back duplicate hour
- Leap seconds; unsynced RTCs
- Offline nodes: re-sync on reconnect

Time Synchronization Hierarchy



Best Practices

- Enforce hourly sync for controllers with poor RTCs
- Use PTP where temporal ordering is critical
- Store both device and server receipt timestamps

Security Considerations

- Protect NTP with ACLs and authentication
- No direct public NTP from OT devices
- Monitor for sudden clock changes

Metadata and Tagging (Semantics)

Why Semantics Matter

- Discoverability - find all temperature sensors in a zone
- Automated joins - connect points across systems
- Reusable analytics - apply rules to all AHUs
- Digital twins - link physical and virtual elements

Tagging Frameworks

Project Haystack:

ahuequipdischargeairtempsensorunit:°C

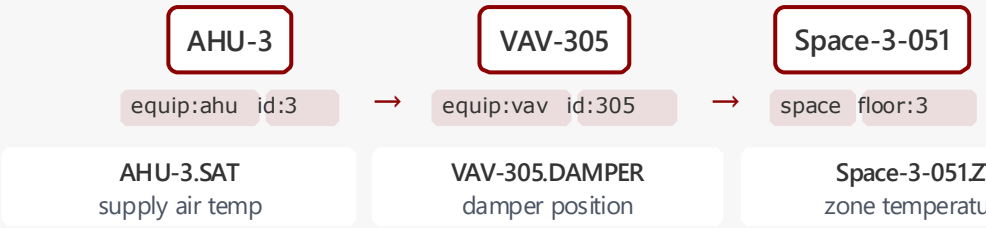
Brick Schema:

Brick:Air_HandlerBrick:Supply_Air_Temperature_Sensor

Minimum Metadata Set

- Equipment type (e.g., AHU, VAV, pump)
- Location (site, building, floor, space)
- Point role (sensor, setpoint, command)
- Engineering units and calibration info
- Relationships (feeds, isPartOf, controls)
- Manufacturer, model, serial numbers

Equipment Relationship & Tagging Example:



Governance

- Controlled vocabulary - avoid "AHU" vs "Air Handler"
- Tag linting - automated validation of metadata
- Code reviews for mappings and transformations
- Versioned dictionaries with change logs

Storage and Access

- Attach tags to points and equipment objects
- Store in graph database or JSON columns
- Cross-reference with BIM (IFC GUIDs)
- Link to CMMS asset IDs for maintenance
- Use stable unique IDs throughout lifecycle

Data Quality and Validation

Core Checks

- Range bounds (hard/soft limits for each sensor type)
- Rate-of-change (detect unrealistic value jumps)
- Spike/outlier detection with statistical methods
- Stuck-at detection (low variance over time window)

Physics/Logic Constraints

- MAT ≈ mix(RAT, OAT, damper) within tolerance
- CHW ΔT vs valve% and load consistency
- Fan speed vs duct static relationship
- Valve position vs temperature response patterns

Process Awareness

- Mode-conditional thresholds (occupied vs unoccupied)
- Maintenance flags suppress false alarms during work
- Context-based validation (weather, schedules, seasons)

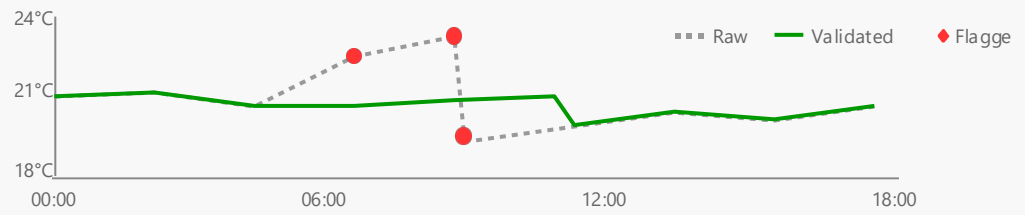
Pipeline Design

- Edge pre-checks + cloud validation layers
- Quarantine bad data without discarding
- Quality flags propagated and stored with data
- Never overwrite raw data; store validated separately

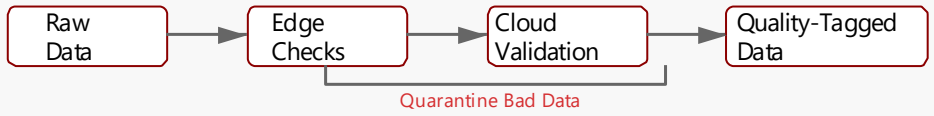
Calibration & Drift

- Reference sensors/periodic checks vs standards
- Apply calibration offsets with audit trail
- Alert on long-term bias vs peers or benchmarks
- Calibration history tracked in metadata

Trend Plot: Raw vs Validated Data with Flagged Intervals



Validation Pipeline Flow:



Missing Data Handling

Gap Types & Causes

- Network outages, sensor faults, scheduled maintenance
- DST shifts, power failures, communication errors
- Controller resets, buffer overflows

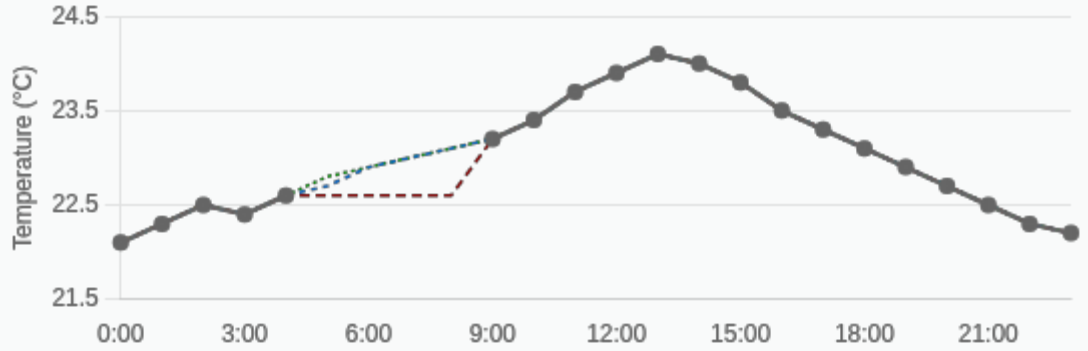
Filling Strategies

- **Forward/backward fill**: States and slow-changing variables (with TTL)
- **Linear/spline interpolation**: Short gaps in continuous variables
- **Model-based/Kalman**: Smoother estimates for correlated points
- **Seasonal baselines**: For energy/occupancy patterns
- **No fill**: Leave gaps for fault analysis (preferred for FDD)

Aggregations with Gaps

- Window completeness thresholds (≥80% samples required)
- Weighted/partial aggregates with uncertainty markers
- Gap-aware statistics (min/max/avg with coverage %)

Time-Series Gap Filling Example:



Forward Fill: Holds last valid value until TTL expires

Linear Interpolation: Draws straight line between valid points

Model-Based: Uses correlations with other points

No Fill: Preferred for fault detection analysis

Business Rules

- Billing/compliance requires documented estimation
- FDD prefers gaps to avoid masking faults
- Flag all imputations in reports/dashboards

Edge Cases

- Binary signals: use bounded forward-fill with TTL
- DST: explicit handling of missing/duplicate hour
- Power outages: restore strategy with priorities

Cybersecurity

Cybersecurity Fundamentals

For Building Systems

Core Principles

- CIA triad (Confidentiality, Integrity, Availability)
- Least privilege access control
- Defense-in-depth layering
- Zero-trust network mindset

Risk & Threat Modeling

- Identify assets, trust boundaries, data flows
- STRIDE-style threat analysis
- High-impact scenarios: unauthorized control, data exfiltration, ransomware

Secure-by-Default

- Disable unused services/ports
- Strong passwords/keys
- MFA where feasible
- Patch and vulnerability management

Cryptography & Secrets

- TLS everywhere; certificate lifecycle
- Hardware roots (TPM/HSM)
- Secrets management (vaults)
- No credentials in code/configs

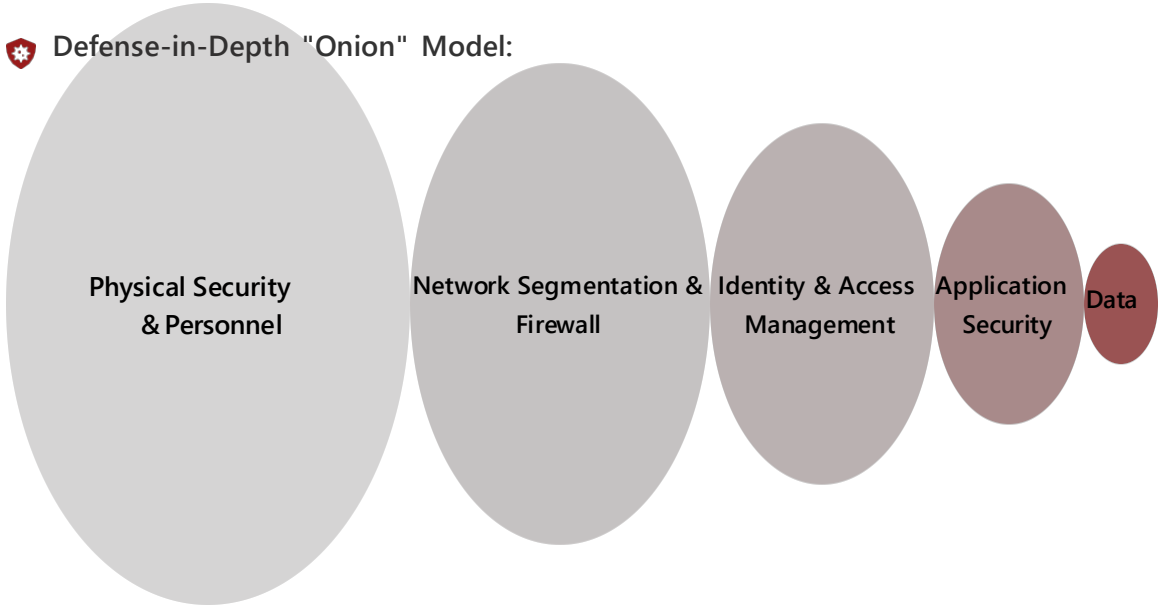
Governance & Culture

- Asset inventory, baselines, SBOMs
- Vendor security requirements
- Policies, training, phishing awareness
- Third-party risk management

Monitoring & Recovery

- Centralized logs, anomaly detection
- Tested backups and restore runbooks
- Incident response playbooks
- Periodic table-top exercises

Defense-in-Depth "Onion" Model:



BAS-Specific Security Controls:

- Air-gapped OT networks or secure DMZ with strict firewall rules
- Role-based access control for BAS interfaces & controllers
- Controlled change management for BAS programming
- Secure remote access with logging, MFA, and time-limited sessions

Session Agenda and Time Plan

Part A: BAS fundamentals

50 min

- BAS architecture
- Core subsystems
- Control logic strategies
- Common protocols

Part B: IT & data

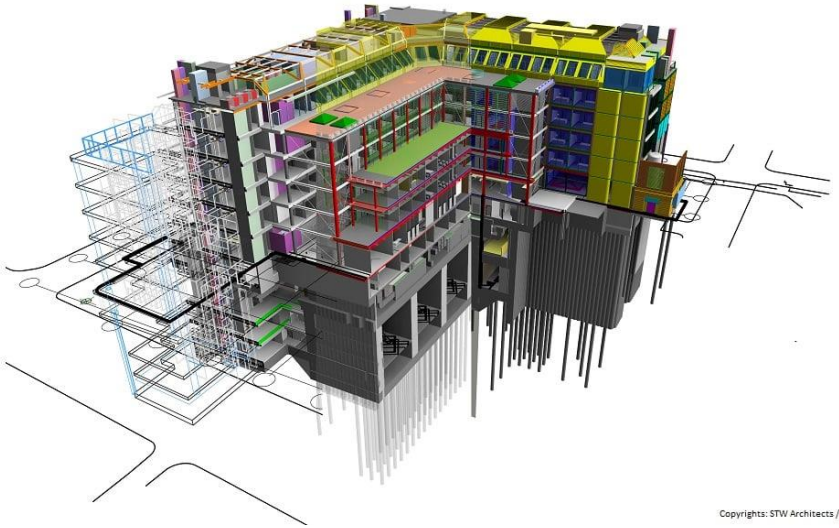
50 min

- Network infrastructure
- Edge→cloud architecture
- Time-series data
- Cybersecurity controls

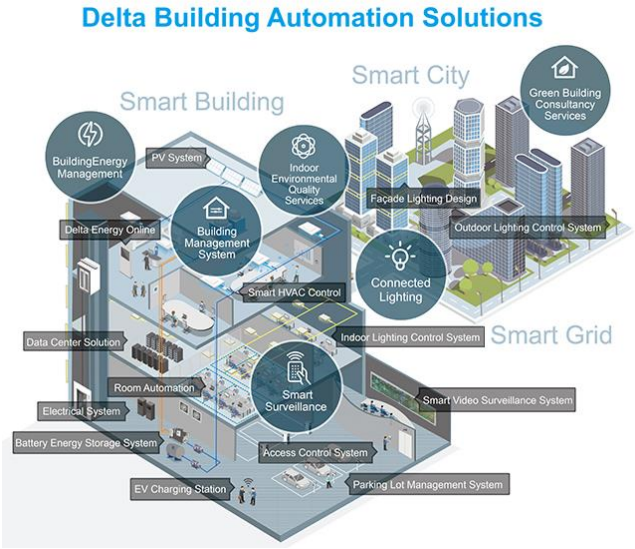
Part C: BIM

50 min

- Data-rich BIM models
- IFC/COBie standards
- Digital twin integration
- BIM-to-BAS mapping



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Activities

10 min

- Interactive polls
- Data tagging mini-task
- Short reflection

Q&A

5 min

- Open questions
- Next week preview

BIM & Building Informatics

What Is BIM? (Building Information Modeling)

Definition

- A data-rich, object-oriented digital representation of a facility
- Combines geometry + properties + relationships
- Not just graphics, but a complete database of building information

BIM vs "a 3D model"

- Parametric objects (walls, ducts, equipment), not just geometry
- Structured properties enable queries, automation, analytics
- Contains relationships between elements (spatial, logical, systems)
- Changes propagate across the entire model (smart objects)

Lifecycle Value

- Early-phase: clash detection, design coordination
- Construction: 4D scheduling, cost estimation (5D)
- Operations: performance monitoring, maintenance (6D)
- Handover to Facility Management: asset data, system topology, space info

For Building Informatics

- Spatial context for sensors and equipment
- Topology (systems, flows) for FDD and impact analysis
- Foundation for digital twins and live data overlays
- Structure for connecting asset management and BAS

BIM Model Visualization:



Floor 3 - HVAC Systems View

BIM Query Example:

```
SELECT Equipment, Location, Systems FROM
BIMModel
WHERE EquipmentType = 'AHU' AND
Floor = '3'
```

Result: AHU-03
Location: Mechanical Room 3-042
Systems: HVAC, Zone 3-East
Serves: Rooms 3-050 through 3-065

BIM vs CAD

Understanding fundamental differences and operational implications

CAD Characteristics

- **Geometry-centric:** 2D/3D drafting with limited semantic data
- **Manual coordination:** Changes require updates across multiple drawings
- **File-based:** Separate DWG files for plans, sections, elevations
- **Static information:** Symbols with minimal embedded metadata
- **Limited analysis:** Mainly visual reference, not computational

BIM Characteristics

- **Object-parametric:** Intelligent objects with constraints & relationships
- **Coordinated views:** Changes propagate across all views automatically
- **Database-driven:** Single source of truth for geometry & data
- **Rich metadata:** Properties embedded in objects for downstream use
- **Analysis-ready:** Supports simulation, costing, scheduling

Collaboration

- Multi-discipline coordination (architectural, structural, MEP)
- Clash detection and issue tracking
- Version control and change management

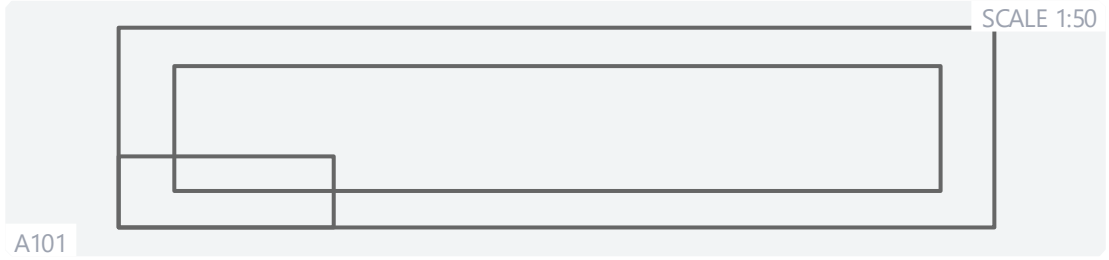
Deliverables

- Models (discipline/federated)
- Drawings, schedules, quantities
- Data exports (IFC, COBie)

Operations Impact

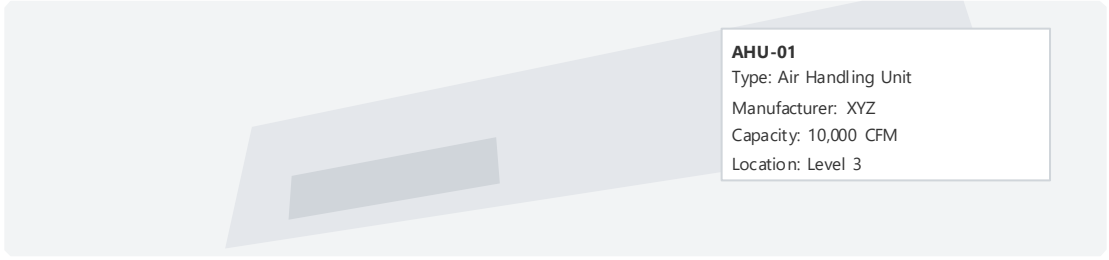
- Reliable asset lists & system topology
- Easier updates and handover
- Integration with CMMS, IoT, analytics

CAD Drawing (2D Lines)



Lines without intelligence or data

BIM Model (3D + Data)



Objects with properties & relationships


Hospitals
Offices
Resorts
Retail
Residential
Education
Data Centers
Infrastructure
...
Detailed Design
Structural Analysis
Human Analysis
Solar Analysis
BIM
(Building Information Modeling)
is used for:
Clash Detection
Quantity Takeoff
Logistics
4D Simulation
Marketing
Facilities Management
Procurement
Lower costs
33%
Faster delivery
50%
Lower emissions
50%
Improvement in exports
50%
Virtual Reality
Mixed Reality
Augmented Reality

IFC and COBie (Data Standards)

Open data standards for building information exchange

IFC (Industry Foundation Classes)

An **open, standardized data format** used for exchanging building and construction data across different software platforms.

- **Developed by**: **buildingSMART International**
- **Purpose**:
  - Enables **interoperability** between different BIM tools.
  - Describes both the **geometry** (3D) and **properties** of building elements.
- **File formats**: .ifc, .ifczip, .ifcXML, etc.
- **Includes data such as**:
  - Elements like walls, doors, pipes, ducts, cable trays
  - System connections and spatial relationships

Best suited for: **Design**, **clash detection**, **construction coordination**, and **facility operations**

COBie (Construction-Operations Building info exchange)

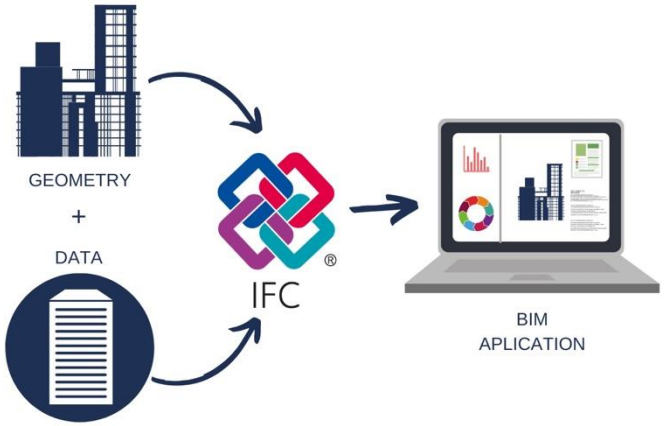
A structured data format used to **deliver asset information** from design and construction teams to **facility managers** at the end of a project.

- **Format**: Typically an **Excel spreadsheet**, but also supported in XML or database formats.
- **Contents include**:
  - Equipment and asset data (e.g., serial numbers, manufacturers)
  - Space and zone information
  - Maintenance schedules, warranty info, contact details
- **Origin**: Developed from U.S. military projects; now supported in **ISO 19650** standards.

Best suited for: **Project handover → Facility Management (FM)**

Comparison: IFC vs COBie

Feature	IFC	COBie
Type	3D model data format	Tabular asset data format
Structure	Object-oriented (geometry + properties + relationships)	Structured tables (assets, spaces, systems)
Purpose	Data exchange across all project phases	Data handover to FM
Format	.ifc / .ifczip / .ifcXML	.xlsx, .xml, databases
Use case	Interoperability between BIM tools	BIM-to-FM transition
Developed by	buildingSMART International	US Army / buildingSMART / ISO 19650



https://www.buildingsmart.org/standards/bsi-standards/industry-foundation-classes/

Revit Families and Parameters (for MEP/FM Readiness)

Family Types

- **System families**: walls, ducts, pipes (host-based, template-defined)
- **Loadable families**: equipment, terminals, fixtures (portable .rfa files)
- **In-place**: custom one-off instances (use only when necessary)

Parameters

- **Type vs instance**: shared across all instances vs unique to each placement
- **Parameter types**: built-in (Revit standard), project (local), shared (global GUID)
- **Shared parameters**: enable consistent schedules/exports, maintain GUIDs

MEP Connectors

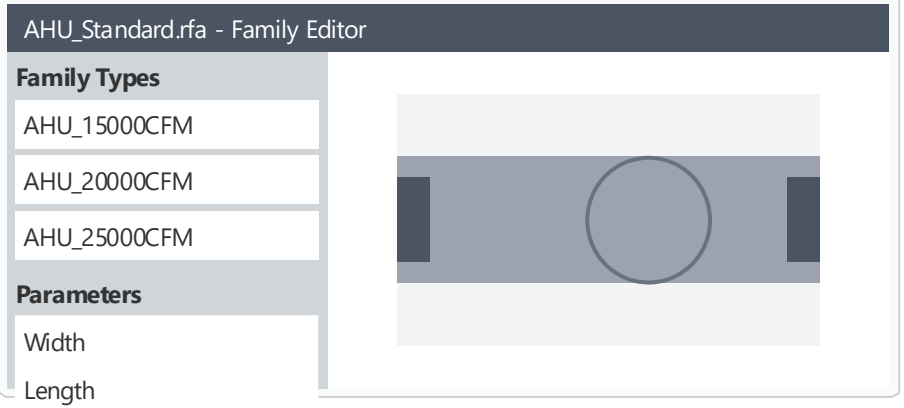
- Define **flow direction**, **system type** (CHW, HHW, SA, RA), and **sizes**
- Maintain **system connectivity** for network/circuit analysis
- Support **pressure/flow calculations** and **clash detection**

Data Extraction

- **Schedules and filters**: visibility control, counting, verification of data completeness
- **APIs**: Dynamo/Forge for automation, parameter mapping, bulk updates
- **Exports**: Parameter mapping to IFC Property Sets and COBie fields for handover



Revit Family Editor - AHU with Parameters



Key FM Parameters for AHU:

AssetID | Manufacturer | Model | SerialNumber | Flow_Rate | Power_Load | ServingZones | MaintenanceFreq | FilterType

BIM as a Data Source for Informatics

Extractable Entities

- Spaces/zones with area, volume, type attributes
- Systems (HVAC, electrical, plumbing) with parameters
- Equipment with manufacturer, model, capacity details
- Terminals (diffusers, outlets) with flow rates
- Meters and monitoring points for telemetry binding

Useful Relationships

→ **Served By/Feeds**: Equipment → Terminal → Space
Example: AHU-3 feeds VAV-305 serves Room-3-051

Part Of: Equipment → System
Example: Fan coil unit is part of HVAC-System-3

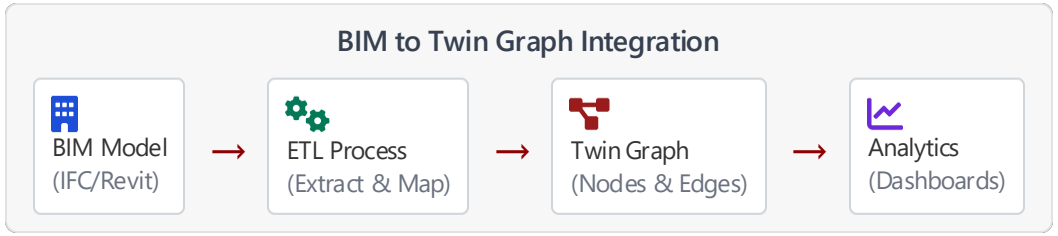
Located In: Space/Floor/Building hierarchy
Example: VAV-305 located in Mechanical Room 3-02

Exports and Formats

- **IFC** (IFC4 recommended) for open standard exchange
- **CSV schedules** for tabular equipment and space data
- **JSON via APIs** for direct integration with platforms
- **COBie** formatted spreadsheets for asset/maintenance

Mapping to Operations

- Crosswalk BIM GUIDs to CMMS Asset IDs and BAS point tags
- Build a "twin graph" to unify BIM semantics with live telemetry
- Store mapping tables in database for persistent relationships
- Enable bidirectional updates: changes in BIM → operations



Linking BIM to Sensor Data

Identity and Binding

Identifier Strategy

- Stable unique IDs: IFC GUIDs, Revit UniqueId, or enterprise Asset ID
- Physical labels/QR/NFC tags in the field linking to digital ID
- Persistent identification across lifecycle changes

Binding Models

- Equipment-level: link AHU in BIM to AHU device in BAS (device_id)
- Point-level: link SAT sensor object to BAS point(s) and unit metadata
- Spatial linkage: point → equipment → space(s)

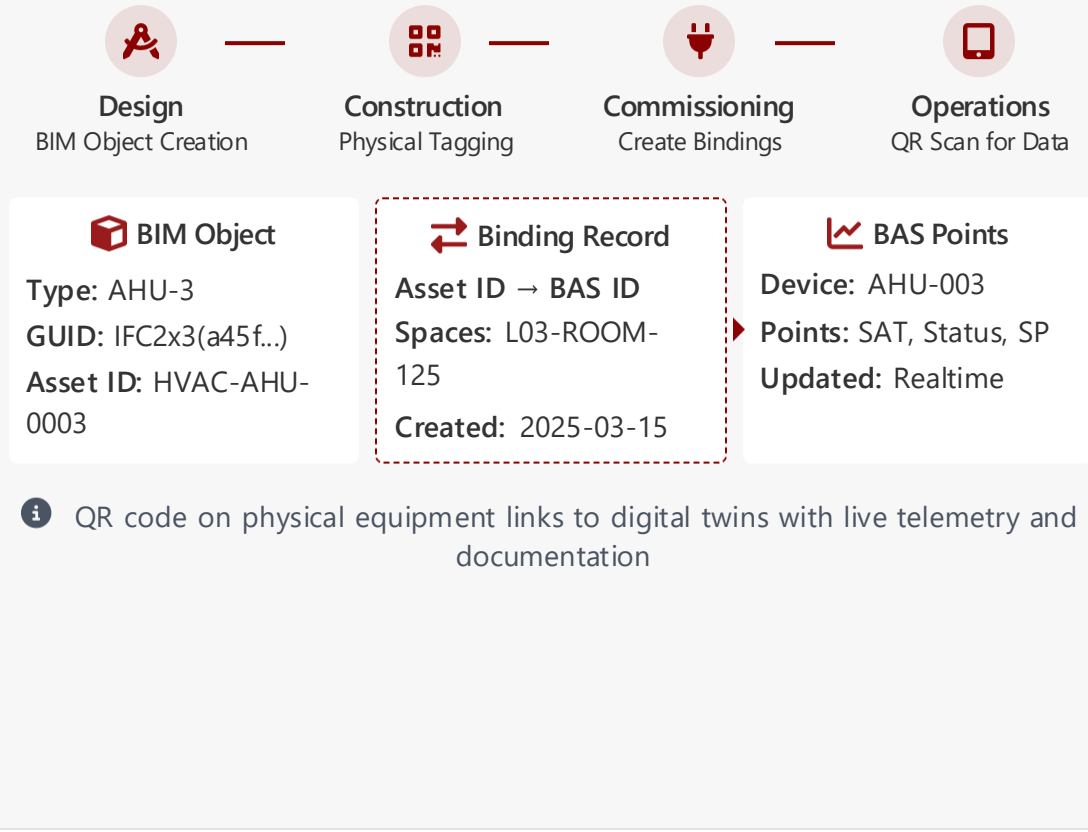
Data Alignment

- Units and coordinate systems; orientation/handedness
- Naming crosswalks for consistent references
- Versioning: model revisions vs operational changes

Governance

- Approval workflow for creating/changing bindings
- Automated validation (e.g., reconcile equipment counts, check missing links)
- Change management process for handling equipment replacements

Physical-Digital Lifecycle Binding



Spatial Mapping & Coordinates

From Models to Reality

Coordinate Systems

- Revit: Project Base Point vs Survey Point; Shared Coordinates
- IFC: local placement hierarchies; Coordinate Reference System mapping
- Managing multiple coordinate systems across software platforms

Georeferencing

- Align to a site CRS (e.g., EPSG code) for multi-building campuses
- Tie-ins for GIS integration and outdoor-to-indoor transitions
- Common coordinate reference systems and transformations

Indoor Positioning

- Define floor/zone/space identifiers; level elevations and z-order
- IPS technologies: BLE beacons, Wi-Fi RTT, UWB anchors for RTLS
- Standard naming conventions: L03, Space-3-051 for clarity

Practical Alignment

- As-built verification (laser scans/point clouds) to reduce drift
- Tolerance policies: acceptable deviations, update cycles
- Documentation of coordinate systems for future reference

Integrating BIM with GIS: Lessons Learned and Best Practices



Site georeferencing with survey control points

https://resources.esri.ca/education-and-research/integrating-bim-cad-with-gis-lessons-learned-and-best-practices

Building Level Stacking

Level 04 (z=16.500m)
Level 03 (z=12.000m)
Level 02 (z=8.000m)
Level 01 (z=4.000m)
Ground (z=0.000m)

Digital Twins & BIM-IoT Integration

Digital Twin: Definitions & Layers

Concept

- Virtual representation of physical assets with data synchronization
- Bridges physical-digital gap through continuous feedback

Layers

1. **Geometry:** 3D model (BIM/scan)
2. **Semantics:** Equipment/space types, relationships (graph)
3. **Telemetry:** Time-series, events, states
4. **Analytics:** KPIs, FDD, predictions
5. **Workflow:** Work orders, procedures, human-in-the-loop

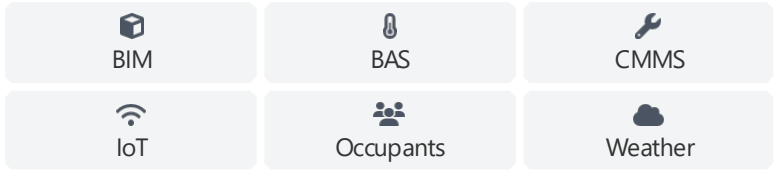
Capabilities

- Real-time monitoring in context, what-if simulations
- Asset genealogy and version tracking
- Topology-aware impact analysis and root-cause tracing
- Decision support and automated workflows

Non-goals

- Not necessarily photorealistic; value comes from data+semantics
- Not a single product but an architecture connecting systems

Data Sources

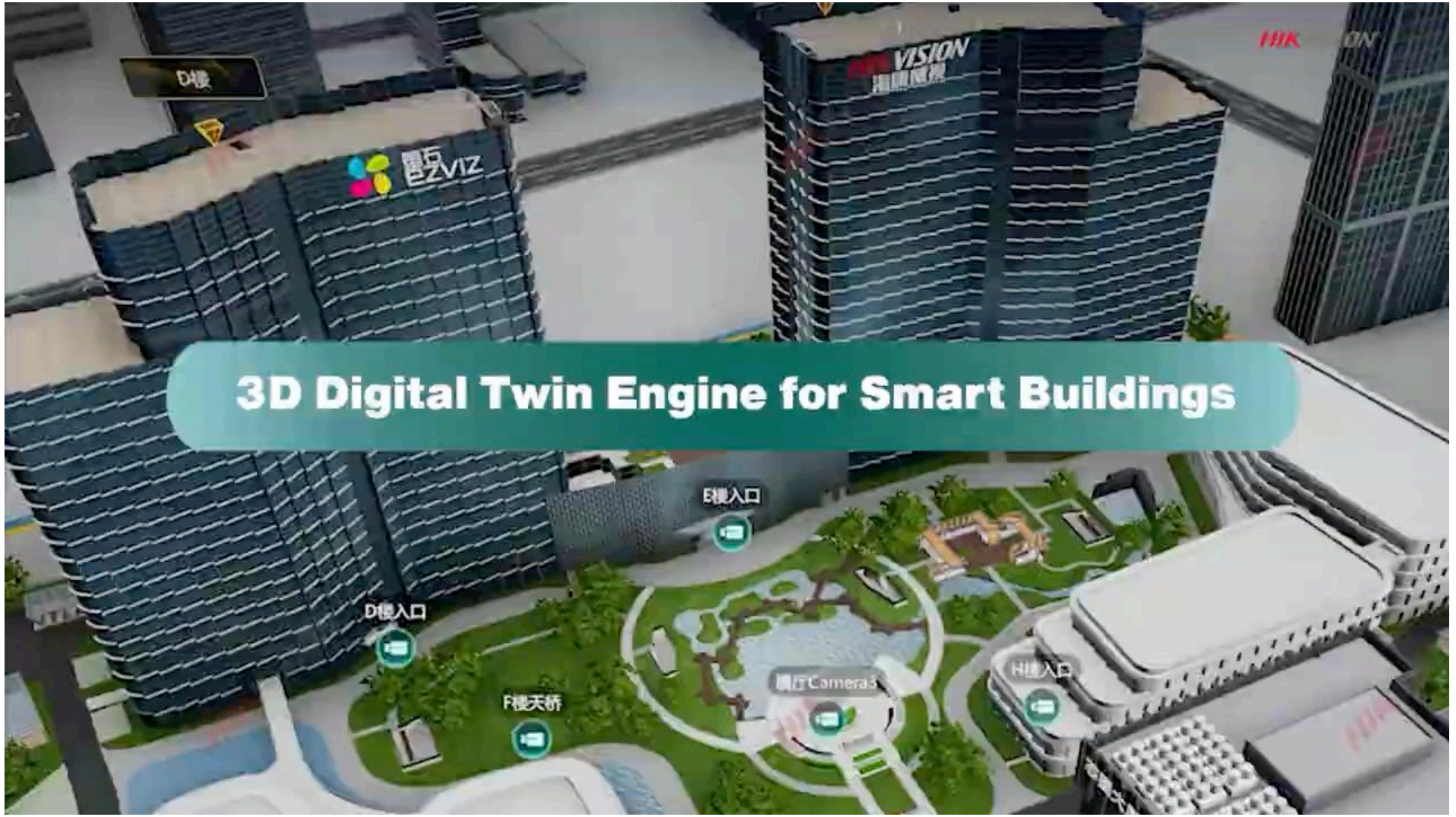


Source	Example
BIM	Building layout and geometry
BAS	HVAC, lighting, alarms
CMMS	Maintenance schedules, work orders
IoT	Sensors, meters, wearables
Occupants	Booking, access, feedback
Weather	Forecasts, outdoor conditions



https://www.12dsynergy.com/innovation-showcase/digital-twins-explained/

3D Digital Twin Engine for Smart Buildings



https://www.youtube.com/watch?v=2QH8wVlcEoI

BIM–IoT Integration Workflow

ETL (Data Extract, Transform, Load) to Experience

BIM–IoT integration combines Building Information Modeling's digital 3D models with real-time data from Internet of Things (IoT) sensors to create a dynamic, intelligent ecosystem for buildings and construction projects.

Source alignment

- Export IFC/COBie/CSV
- Pull BAS point lists
- CMMS asset register
- Define mapping rules

ETL & binding

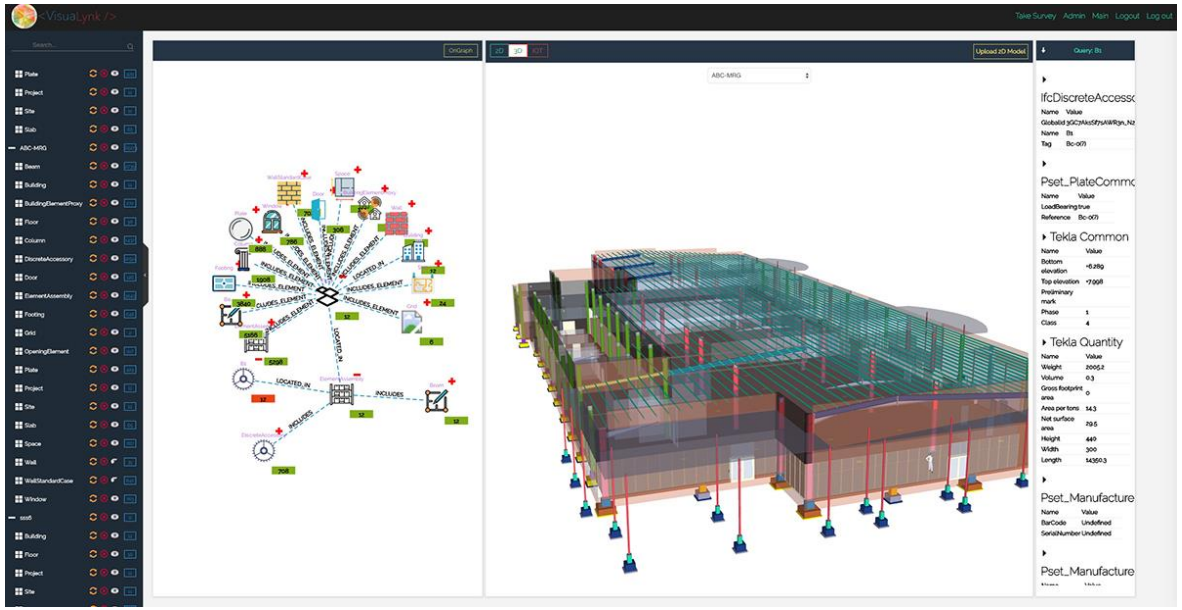
- Load entities & relations
- Attach tags (Haystack/Brick)
- Create crosswalks (GUID — asset_id ↔ point_id)

Bind live data

- Subscribe to MQTT/OPC/BACnet
- Normalize timestamps/units
- Materialize KPIs and alarms

Apps & views

- 2D/3D viewers & dashboards
- Topology explorers & FDD
- Mobile: QR → live status → work order



Connecting IoT Data to BIM

https://aec-business.com/connecting-iot-data-to-bim/

Use Case: BIM-Driven Fault Isolation (End-to-End)

Alert Scenarios

Low ΔT in Chilled Water Plant

Poor temperature differential between supply and return water indicating inefficient energy usage

Zone Overheating on Level 3

Multiple spaces reporting temperatures above setpoint despite cooling demand

Twin-Assisted Steps

1. Identify Affected Assets
Use BIM topology to locate equipment and impacted spaces

2. Trace Relationships
Follow upstream/downstream connections (valves, coils, AHUs)

3. Pull Telemetry
Retrieve relevant time-series data for correlated analysis

4. Apply Rules & SoO
Compare actual vs expected behavior using FDD rules

Visualization: Zone Overheating Investigation



Root Causes Ranked

1. **Sensor Bias (83%)**
   SAT sensor reading incorrect
2. **Valve Stuck (67%)**
   Chilled water valve position
3. **Controller Issue (45%)**
   PID tuning parameters

Impact Assessment

- 8 affected spaces on Level 3
- 22 occupants impacted
- Est. energy penalty: 12.4 kWh/day
- Comfort violation: 3.2 hours

CMMS Integration

Auto-Generated Work Order

Priority:	Medium
Location:	AHU-3, Level 3 East
Task:	Verify SAT sensor calibration
Parts:	Temperature sensor (optional)

Integration Challenges (Reality Check)

Model Fidelity

- Incomplete/incorrect BIM data; missing systems or as-builts
- Drifts over time without change control

Semantic Gaps

- Vendor naming inconsistencies; unit mismatches
- Ambiguous IDs; missing relationships

Technical Hurdles

- Gateway/protocol idiosyncrasies; latency issues
- 3D model performance in web/mobile
- High-cardinality data; permissions complexity

Organizational Barriers

- Ownership of data; vendor lock-in
- Fragmented responsibilities across teams
- Privacy and compliance concerns (location/occupancy)

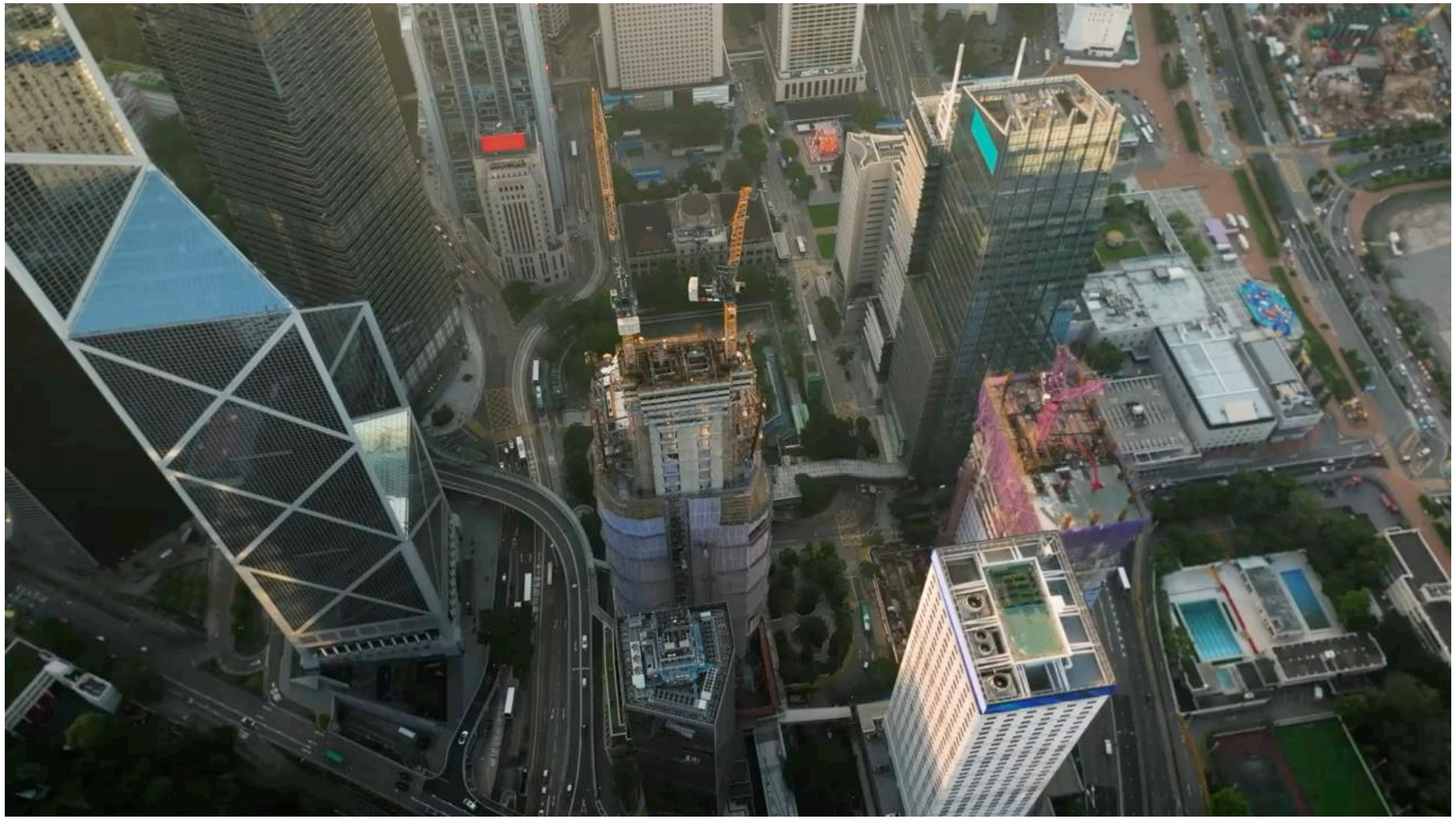
Mitigations

- Governance: data dictionaries, RACI, change management
- Validation pipelines; incremental rollout
- Open standards adoption (IFC, COBie, Haystack/Brick)
- Clear SLAs with vendors and data quality contracts

Top Challenges with Mitigations:

Inconsistent BIM-BAS naming conventions Different naming for same equipment across systems	→	**Crosswalk dictionary + validation rules** Automated validation with mapping tables
BIM model outdated vs. actual building Renovations/changes not reflected in model	→	**Change management workflow** FM approval process with model updates
Data privacy concerns with occupancy Personal movement tracking through BAS/BIM	→	**Aggregation & anonymization policies** Zone-level vs. individual tracking; retention limits

What is a Digital Twin in Construction? | Explained in Under 5 Minutes



https://www.youtube.com/watch?v=3BB1KbfluwE

Thanks
&
Any Question?